

COURSE NOTES

Abstract Algebra III

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Chapter 1

Lecture 1: Modules

Message: QQ, BB. HW 30%, Mid 20%, Final 50%

1.1 Basic Definitions and Examples

Definition 1.1.1. Let R be a ring with 1, and let M be an abelian group. We say that M is a left R -module if R “left acts” on M :

$$R \times M \rightarrow M, \quad (r, m) \mapsto rm$$

such that

$$\begin{aligned} r(m + n) &= rm + rn, \\ (r + s)m &= rm + sm, \\ (rs)m &= r(sm), \\ 1m &= m. \end{aligned} \quad \text{for any } r, s \in R, m, n \in M$$

A **right R -module** is defined by $(r, m) \mapsto mr$ with the axiom $m(rs) = (mr)s$. If R is commutative, left-modules and right-modules coincide. Otherwise, one can transform using the opposite ring R^{op} .

Consider M an R -module, then $\forall r \in R$, $\varphi_r : M \rightarrow M$ by $\varphi_r(m) = rm$ is an abelian group homomorphism. Let

$$\text{End}(M) = \{\varphi : M \rightarrow M \text{ is a abelian group homomorphism} \}$$

Then $\text{End}(M)$ is a ring with composition as the multiplication.

Theorem 1.1.2. Let M be an abelian group. Then M is an R -module if and only if there is a ring homomorphism $R \rightarrow \text{End}(M)$.

Proof. ($\Rightarrow$) Suppose M is an R -module. Define $\rho : R \rightarrow \text{End}(M)$ by

$$\rho(r)(m) = rm, \quad \forall r \in R, m \in M.$$

For each $r \in R$, the map $\rho(r) : M \rightarrow M$ is a group homomorphism since $r(m_1 + m_2) = rm_1 + rm_2$. Moreover, ρ is a ring homomorphism:

- $\rho(r + s)(m) = (r + s)m = rm + sm = \rho(r)(m) + \rho(s)(m)$, so $\rho(r + s) = \rho(r) + \rho(s)$.
- $\rho(rs)(m) = (rs)m = r(sm) = \rho(r)(\rho(s)(m))$, so $\rho(rs) = \rho(r) \circ \rho(s)$.
- $\rho(1)(m) = 1m = m$, so $\rho(1) = \text{id}_M$.

($\Leftarrow$) Suppose $\rho : R \rightarrow \text{End}(M)$ is a ring homomorphism. Define the scalar multiplication $R \times M \rightarrow M$ by

$$(r, m) \mapsto \rho(r)(m).$$

We verify the module axioms:

- $r(m_1 + m_2) = \rho(r)(m_1 + m_2) = \rho(r)(m_1) + \rho(r)(m_2) = rm_1 + rm_2$.
- $(r + s)m = \rho(r + s)(m) = (\rho(r) + \rho(s))(m) = \rho(r)(m) + \rho(s)(m) = rm + sm$.
- $(rs)m = \rho(rs)(m) = (\rho(r) \circ \rho(s))(m) = \rho(r)(\rho(s)(m)) = r(sm)$.
- $1m = \rho(1)(m) = \text{id}_M(m) = m$.

Hence M is an R -module. □

Example 1.1.3. (1) A vector space is a module over a field.

(2) An abelian group is a module over the ring of integers $\mathbb{Z}$.

$$\mathbb{Z} \times M \rightarrow M, \quad (n, m) \mapsto n \cdot m := m + m + \cdots + m \quad (n \text{ times})$$

(3) An ideal I of a ring R is a module over R .

(4) Let M be an abelian group. Then M is an $\text{End}(M)$ -module via

$$\text{End}(M) \times M \rightarrow M, \quad (\varphi, m) \mapsto \varphi(m).$$

(5) Let V be a vector space over a field K , and let $A : V \rightarrow V$ be a linear transformation. Then V is a $K[x]$ -module via

$$K[x] \times V \rightarrow V, \quad (f(x), v) \mapsto f(A)v.$$

(6) If $|M| = 1$, then M is called the zero module.

(7) Let R be a ring with 1 and $A \in M_n(R)$. The solution of $AX = 0$ where $X \in R^n$ is a right R -module.

Definition 1.1.4 (R -submodule). Let R be a ring (with 1) and let M be a left R -module. A subset $N \subseteq M$ is called an R -submodule of M if:

1. $0 \in N$;
2. $n_1, n_2 \in N \implies n_1 + n_2 \in N$;

3. $n \in N \implies -n \in N$;
4. $r \in R, n \in N \implies rn \in N$.

Equivalently, N is a subgroup of the abelian group $(M, +)$ that is closed under the R -action.

Example 1.1.5. (1) Let V be a vector space over a field K , then any subspace of V is a k -submodule of V .

(2) Let G be an abelian group, then any subgroup of G is a $\mathbb{Z}$ -submodule of G .

(3) Let R be a ring, ${}_R R$ is a left (regular) R -module. Then any left ideal of R is a R -submodule of ${}_R R$.

(4) Let G be an abelian group, which is an $\text{End}(G)$ -module. Then any full invariant subgroup of G is a $\text{End}(G)$ -submodule of G .

Question: Are submodules of a finitely generated R -module finitely generated?

Answer: No, not in general. This property holds if and only if the ring R is Noetherian.

Example 1.1.6 (Submodule of a f.g. module that is not f.g.). Let k be a field and consider the polynomial ring $R = k[x_1, x_2, x_3, \dots]$ in infinitely many variables.

- The ring R itself can be viewed as a left R -module, which we denote by $M = R$. M is finitely generated, for instance by the single element $1 \in R$.
- Consider the ideal $I = \langle x_1, x_2, x_3, \dots \rangle$ generated by all the variables. This ideal I is a submodule of $M = R$.
- This submodule I is not finitely generated. To see this, suppose for contradiction that I is finitely generated by a set of polynomials $\{f_1, \dots, f_n\}$. Let N be the largest index of any variable appearing in any of the polynomials $f_1, \dots, f_n$. Then any element in the ideal generated by $\{f_1, \dots, f_n\}$ must be a polynomial in variables $x_1, \dots, x_N$. However, the polynomial x_{N+1} is in I but cannot be expressed as a combination of $f_1, \dots, f_n$. This is a contradiction.

Therefore, the ideal I is a submodule of the finitely generated R -module R which is not itself finitely generated. This shows that the ring R is not Noetherian.

Definition 1.1.7. Let M be an R -module, $M_i, i \in I$ are R -submodules of M . Then

$$\bigcap_{i \in I} M_i$$

is an R -submodule of M .

$$\sum_{i \in I} = \left\{ \sum_{i \in I \text{ finite}} m_i \mid m_i \in M_i \right\}$$

is an R -submodule of M .

Definition 1.1.8. Let M be an R -module, $S \subseteq M$ a subset.

$$\langle S \rangle = \left\{ \sum_{\text{finite}} a_i s_i \mid a_i \in R, s_i \in S \right\}$$

or

$$\langle S \rangle = \bigcap_{\substack{N \text{ is an } R\text{-submodule of } M, \\ S \subseteq N}} N$$

is an R -submodule of M , called the R -submodule of M generated by S .

If $M = \langle S \rangle$, then we say M is generated by S . In particular, if $|S| = 1$ then we call M a cyclic module. If $|S| < \infty$ then we call M a finite generated module.

Definition 1.1.9. Let R be a ring and let M be a left R -module. We say that M is a Noetherian module if every ascending chain of submodules

$$M_1 \subseteq M_2 \subseteq M_3 \subseteq \cdots$$

stabilizes; that is, there exists $n \geq 1$ such that

$$M_n = M_{n+1} = M_{n+2} = \cdots.$$

Equivalently, M is Noetherian if every submodule of M is finitely generated.

1.2 Quotient Modules and Homomorphism Theorems

Definition 1.2.1. Let R be a ring, let M be a left R -module, and let N be a submodule of M . The quotient module (or factor module) of M by N is the quotient abelian group

$$M/N = \{x + N \mid x \in M\},$$

with scalar multiplication defined by

$$r(x + N) = rx + N \quad (r \in R, x \in M).$$

This is well-defined, and with this operation M/N becomes a left R -module.

Example 1.2.2. (1) Let R be a ring and an R -module, I be an ideal of R , then R/I is a quotient R -module of R by I .

(2) Consider the ring $\mathbb{Z}$ and the module $\mathbb{Z}$ over $\mathbb{Z}$, then for $n \in \mathbb{Z}$, $n\mathbb{Z}$ is a submodule of $\mathbb{Z}$, and $\mathbb{Z}/n\mathbb{Z}$ is a quotient $\mathbb{Z}$ -module of $\mathbb{Z}$ by $n\mathbb{Z}$. Furthermore, $\mathbb{Z}/n\mathbb{Z}$ is a cyclic module.

Definition 1.2.3. Let R be a ring, and let M and N be left R -modules. A group homomorphism

$$f : M \rightarrow N$$

is called an R -module homomorphism (or R -linear map) if for all $x, y \in M$ and all $r \in R$, one has

$$f(rx) = rf(x) \quad \forall r \in R, x \in M.$$

Definition 1.2.4. Let f be an R -module homomorphism. The kernel of f is the set

$$\ker(f) = \{x \in M \mid f(x) = 0\}.$$

$\ker(f)$ is a submodule of M . The image of f is the set

$$\operatorname{im}(f) = \{f(x) \in N \mid x \in M\}.$$

$\operatorname{im}(f)$ is a submodule of N .

If $\ker(f) = 0$, then f is injective. If $\operatorname{im}(f) = N$, then f is surjective. If $\ker(f) = 0$ and $\operatorname{im}(f) = N$, then f is bijective.

A bijective R -module homomorphism is called an isomorphism.

Example 1.2.5. Let R be a ring and M, N be R -modules. Then $\operatorname{Hom}_R(M, N)$ is an R -module when R is commutative.

Theorem 1.2.6. If $f : M \rightarrow N$ is an R -module homomorphism, then we have a canonical R -module isomorphism:

$$M/\ker(f) \rightarrow \operatorname{im}(f)$$

Theorem 1.2.7. Let N be an R -submodule of M , $\pi_N : M \rightarrow M/N$ be a quotient map. Then π_N induces a natural bijection:

$$\{R\text{-submodules of } M \text{ containing } N\} \rightarrow \{R\text{-submodules of } M/N\}$$

Theorem 1.2.8. If $N \subseteq H$ are two R -submodules of M , then

$$M/H \cong M/N/(H/N),$$

Theorem 1.2.9. If N, H are R -submodules of M , then

$$\frac{H+N}{N} \cong \frac{H}{H \cap N}$$

Proposition 1.2.10. *Let M be an R -module, N a submodule of M , M' an R -module, and $\varphi : M \rightarrow M'$ a morphism such that $N \subseteq \ker(\varphi)$. Then there exists a unique morphism*

$$\varphi_N : M/N \rightarrow M'$$

such that the following diagram commutes. Moreover, $\ker(\varphi_N) = \ker \varphi / N$.

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_N \downarrow & \nearrow \varphi_N & \\ M/N & & \end{array}$$

Remark 1.2.11. *The pair $(M/N, \pi_N)$ is unique in the following sense: if (S, θ) is a pair such that*

- *S is an R -module,*
- *$\theta : M \rightarrow S$ is a surjective morphism satisfying the property of $(M/N, \pi_N)$ described in the above proposition,*

then there exists a unique isomorphism $\psi : M/N \rightarrow S$ such that the following diagram commutes.

$$\begin{array}{ccc} M & \xrightarrow{\theta} & S \\ \pi_N \downarrow & \nearrow \exists! \psi & \\ M/N & & \end{array}$$

Proof. Since (S, θ) satisfies the same universal property as $(M/N, \pi_N)$, we may apply that property to the canonical projection

$$\pi_N : M \rightarrow M/N.$$

Because $N \subseteq \ker(\pi_N)$, there exists a unique morphism

$$u : S \rightarrow M/N$$

such that

$$u \circ \theta = \pi_N.$$

On the other hand, since $(M/N, \pi_N)$ has the universal property of the quotient by N , and since $N \subseteq \ker(\theta)$, there exists a unique morphism

$$\psi : M/N \rightarrow S$$

such that

$$\psi \circ \pi_N = \theta.$$

We now show that ψ and u are inverse to each other.

First, compute

$$(u \circ \psi) \circ \pi_N = u \circ (\psi \circ \pi_N) = u \circ \theta = \pi_N.$$

But the identity morphism $\text{id}_{M/N} : M/N \rightarrow M/N$ also satisfies

$$\text{id}_{M/N} \circ \pi_N = \pi_N.$$

By the uniqueness part of the universal property of $(M/N, \pi_N)$, it follows that

$$u \circ \psi = \text{id}_{M/N}.$$

Similarly,

$$(\psi \circ u) \circ \theta = \psi \circ (u \circ \theta) = \psi \circ \pi_N = \theta.$$

But the identity morphism $\text{id}_S : S \rightarrow S$ also satisfies

$$\text{id}_S \circ \theta = \theta.$$

By the uniqueness part of the universal property of (S, θ) , we obtain

$$\psi \circ u = \text{id}_S.$$

Therefore ψ is an isomorphism, with inverse u .

Finally, the uniqueness of ψ follows directly from the universal property of $(M/N, \pi_N)$: indeed, ψ is the unique morphism

$$\psi : M/N \rightarrow S$$

such that

$$\psi \circ \pi_N = \theta.$$

Hence there exists a unique isomorphism

$$\psi : M/N \rightarrow S$$

for which the diagram commutes. □

Proposition 1.2.12. *There is a unique isomorphism $\bar{\varphi} : \text{coim } \varphi \rightarrow \text{im } \varphi$ such that the following diagram commutes:*

$$\begin{array}{ccc} M & \xrightarrow{\varphi} & M' \\ \pi_{\ker(\varphi)} \downarrow & & \uparrow \\ \text{coim}(\varphi) & \xrightarrow{\bar{\varphi}} & \text{im}(\varphi) \end{array}$$

Recall:

$$\begin{aligned}\varphi : M &\rightarrow M', \quad R\text{-morphism} \\ \text{coker}(\varphi) &= M' / \text{im}(\varphi) \\ \text{coim}(\varphi) &= M / \ker \varphi\end{aligned}$$

Definition 1.2.13. Let M be an R -module and I an ideal of R . Denote by IM the submodule of M generated by all elements am where $a \in I$, $m \in M$.

$$IM = \left\{ \sum_{\text{finite}} a_i m_i \mid a_i \in I, m_i \in M \right\}$$

Proposition 1.2.14. Let M be an R -module. Then $R \rightarrow \text{End}(M)$ is a ring homomorphism.

1. $R \rightarrow \text{End}(M)$ factors through the natural surjection $R \rightarrow R/I$ if and only if $IM = 0$:

$$\begin{array}{ccc} R & \xrightarrow{\rho} & \text{End}(M) \\ & \searrow \pi_I & \nearrow \exists \rho_I \\ & R/I & \end{array}$$

Where ρ factors through π_I means $\exists \rho_I : R/I \rightarrow \text{End}(M)$ such that the diagram above commutes.

2. The module M/IM is naturally endowed with a structure of R/I -module.
3. Let $\varphi : M \rightarrow M'$ be an R -homomorphism. Then $\varphi(IM) \subseteq IM'$, so φ induces a morphism of R/I -modules

$$M/IM \rightarrow M'/IM'.$$

4. If $M = M_1 \oplus \cdots \oplus M_n$, then $M/IM = M_1/IM_1 \oplus \cdots \oplus M_n/IM_n$.
5. $R^n/IR^n \cong (R/IR)^n$.

Proof. (1) Suppose first that ρ factors through π_I . Then there exists a ring homomorphism

$$\rho_I : R/I \rightarrow \text{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

If $a \in I$, then $\pi_I(a) = 0$, so

$$\rho(a) = \rho_I(0) = 0.$$

Therefore $am = 0$ for every $m \in M$. Since IM is generated by elements of the form am with $a \in I$ and $m \in M$, it follows that

$$IM = 0.$$

Conversely, assume that $IM = 0$. Then for every $a \in I$ and every $m \in M$ we have

$$\rho(a)(m) = am = 0.$$

Thus $\rho(a) = 0$, so $I \subseteq \ker \rho$. Hence ρ factors uniquely through the quotient ring R/I : there exists a unique ring homomorphism

$$\rho_I : R/I \rightarrow \text{End}_{\mathbb{Z}}(M)$$

such that

$$\rho = \rho_I \circ \pi_I.$$

(2) Define an action of R/I on M/IM by

$$(r + I) \cdot (m + IM) := rm + IM.$$

We must show that this is well-defined.

If $r + I = r' + I$, then $r - r' \in I$, so

$$(r - r')m \in IM,$$

hence

$$rm + IM = r'm + IM.$$

If $m + IM = m' + IM$, then $m - m' \in IM$. Since IM is a submodule of M , we have

$$r(m - m') \in IM,$$

so

$$rm + IM = rm' + IM.$$

Thus the action is well-defined. The module axioms follow immediately from the module axioms on M . Therefore M/IM is naturally an R/I -module.

(3) Let $\varphi : M \rightarrow M'$ be an R -module homomorphism. Take any element of IM , say

$$x = \sum_{j=1}^t a_j m_j \quad (a_j \in I, m_j \in M).$$

Then

$$\varphi(x) = \sum_{j=1}^t \varphi(a_j m_j) = \sum_{j=1}^t a_j \varphi(m_j).$$

Since each $a_j \in I$ and each $\varphi(m_j) \in M'$, it follows that

$$a_j \varphi(m_j) \in IM'.$$

Hence $\varphi(x) \in IM'$, and therefore

$$\varphi(IM) \subseteq IM'.$$

Thus we may define

$$\bar{\varphi} : M/IM \rightarrow M'/IM', \quad \bar{\varphi}(m + IM) = \varphi(m) + IM'.$$

This is well-defined: if $m + IM = m' + IM$, then $m - m' \in IM$, so

$$\varphi(m) - \varphi(m') = \varphi(m - m') \in IM',$$

hence

$$\varphi(m) + IM' = \varphi(m') + IM'.$$

Finally, $\bar{\varphi}$ is R/I -linear, since

$$\bar{\varphi}((r + I) \cdot (m + IM)) = \bar{\varphi}(rm + IM) = \varphi(rm) + IM' = r\varphi(m) + IM' = (r + I) \cdot \bar{\varphi}(m + IM).$$

(4) Assume that

$$M = M_1 \oplus \cdots \oplus M_n.$$

We first show that

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

If $x \in IM$, then x is a finite sum of elements of the form am with $a \in I$ and $m \in M$. Writing

$$m = m_1 + \cdots + m_n \quad (m_i \in M_i),$$

we get

$$am = am_1 + \cdots + am_n,$$

and each $am_i \in IM_i$. Hence $x \in IM_1 + \cdots + IM_n$, so

$$IM \subseteq IM_1 + \cdots + IM_n.$$

The reverse inclusion is immediate because each $M_i \subseteq M$, hence each $IM_i \subseteq IM$. Therefore

$$IM = IM_1 + \cdots + IM_n.$$

To see that the sum is direct, suppose

$$x_1 + \cdots + x_n = 0 \quad \text{with } x_i \in IM_i \subseteq M_i.$$

Since

$$M = M_1 \oplus \cdots \oplus M_n$$

is a direct sum, we must have $x_i = 0$ for all i . Thus

$$IM = IM_1 \oplus \cdots \oplus IM_n.$$

Now define

$$\Psi : M_1/IM_1 \oplus \cdots \oplus M_n/IM_n \rightarrow M/IM$$

by

$$\Psi(m_1 + IM_1, \dots, m_n + IM_n) = (m_1 + \cdots + m_n) + IM.$$

This map is well-defined and R/I -linear. It is surjective because every element of M can be written as $m_1 + \cdots + m_n$. It is injective because

$$(m_1 + \cdots + m_n) + IM = IM$$

means that

$$m_1 + \cdots + m_n \in IM = IM_1 \oplus \cdots \oplus IM_n,$$

so each $m_i \in IM_i$. Hence

$$m_i + IM_i = IM_i$$

for all i , and the kernel is trivial. Therefore Ψ is an isomorphism:

$$M/IM \cong M_1/IM_1 \oplus \cdots \oplus M_n/IM_n.$$

(5) Since

$$R^n = R \oplus \cdots \oplus R$$

(n copies), part (4) yields

$$R^n/IR^n \cong R/IR \oplus \cdots \oplus R/IR.$$

Therefore

$$R^n/IR^n \cong (R/IR)^n.$$

This completes the proof. □

1.3 Direct Sums and Free Modules

Definition 1.3.1. *Direct sum and direct product ($\bigoplus_{i \in I} M_i$ & $\prod_{i \in I} M_i$).*

Let $(M_i)_{i \in I}$ be a family of R -modules. The direct product of $(M_i)_{i \in I}$ is the module

$$\prod_{i \in I} M_i = \{\varphi : I \rightarrow \cup_{i \in I} M_i \mid \forall i \in I, \varphi(i) \in M_i\}$$

with the componentwise addition and scalar multiplication. i.e.

$$R \times \prod_{i \in I} M_i \rightarrow \prod_{i \in I} M_i, \quad (r, \varphi) \mapsto (r\varphi(i))_{i \in I}$$

The direct sum of $(M_i)_{i \in I}$ is the module

$$\bigoplus_{i \in I} M_i = \{\varphi \in \prod_{i \in I} M_i \mid \#\{i \in I \mid \varphi(i) \neq 0\} < \infty\}$$

Equal when $|I| < \infty$. In a module category, direct product = product, direct sum = coproduct.

Theorem 1.3.2. Suppose that $M_1, \dots, M_n$ are R -submodules of M in which $M = M_1 + \dots + M_n$, $I = [n]$. Then the followings are equivalent:

1. The map $\varphi : \sum_{i \in I} M_i \rightarrow M$ by $\varphi((m_i)_{i \in I}) = \sum_{i \in I} m_i, m_i \in M_i$ is an isomorphism.
2. Any element of M can be uniquely written as a sum of elements of M_i .
3. 0 can be uniquely written as a sum of elements of M_i . More precisely, $0 = 0 + \dots + 0$.
4. For $i \in I$, $M_i \cap (\sum_{j \neq i} M_j) = 0$.

Proof. We prove

$$(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (4) \Rightarrow (1).$$

(1) $\Rightarrow$ (2). If φ is an isomorphism, then for every $m \in M$ there exists a unique

$$(m_1, \dots, m_n) \in \sum_{i=1}^n M_i$$

such that

$$m = \varphi(m_1, \dots, m_n) = m_1 + \dots + m_n.$$

Hence every element of M can be written uniquely as a sum of elements of the M_i .

(2) $\Rightarrow$ (3). Apply (2) to the element $0 \in M$.

(3) $\Rightarrow$ (4). Fix $i \in I$, and let

$$x \in M_i \cap \sum_{j \neq i} M_j.$$

Then $x \in M_i$, and there exist $m_j \in M_j$ for $j \neq i$ such that

$$x = \sum_{j \neq i} m_j.$$

Therefore

$$0 = x - \sum_{j \neq i} m_j$$

is a decomposition of 0 as a sum of elements from the submodules $M_1, \dots, M_n$. By the uniqueness in (3), this decomposition must be the trivial one. In particular,

$$x = 0.$$

Hence

$$M_i \cap \sum_{j \neq i} M_j = 0.$$

(4) $\Rightarrow$ (1). Since

$$M = M_1 + \dots + M_n,$$

the map φ is surjective. It remains to show that φ is injective.

Suppose that

$$\varphi(m_1, \dots, m_n) = 0,$$

that is,

$$m_1 + \dots + m_n = 0 \quad \text{with } m_i \in M_i.$$

For each $i \in I$, we have

$$m_i = - \sum_{j \neq i} m_j \in \sum_{j \neq i} M_j.$$

Since also $m_i \in M_i$, it follows that

$$m_i \in M_i \cap \sum_{j \neq i} M_j = 0.$$

Thus $m_i = 0$ for all i , so

$$\ker \varphi = 0.$$

Therefore φ is injective, and hence an isomorphism.

This proves that the four conditions are equivalent. □

Definition 1.3.3 (free module). *Let M be an R -module.*

1. $x_1, \dots, x_n \in M$ are linearly independent if

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = 0 \Rightarrow a_1 = a_2 = \dots = a_n = 0.$$

2. M is called free if it is generated by a linearly independent subset. This subset is called a basis of M .

3. If M is an R -module with a basis $x_1, \dots, x_n$, then $M = \langle x_1 \rangle \oplus \dots \oplus \langle x_n \rangle$.

Example 1.3.4. $R \oplus \cdots \oplus R = R^n$.

Theorem 1.3.5. *Let R be a commutative ring with 1. Let M be a finitely generated free R -module. Then any basis of M has the same cardinality. The number of elements in a basis is called the rank of the free module.*

Proof. Let $B = \{x_i\}_{i \in I}$ be a basis of M . We first show that any basis is finite. Since M is finitely generated, choose generators $y_1, \dots, y_n$ of M . Each y_k is a finite R -linear combination of elements of B , so there exists a finite subset $I_0 \subset I$ such that

$$y_1, \dots, y_n \in \sum_{i \in I_0} Rx_i.$$

Hence

$$M = \sum_{k=1}^n Ry_k \subseteq \sum_{i \in I_0} Rx_i \subseteq M,$$

so $M = \sum_{i \in I_0} Rx_i$. If $I \setminus I_0 \neq \emptyset$, pick $j \in I \setminus I_0$. Then

$$x_j \in M = \sum_{i \in I_0} Rx_i,$$

contradicting that B is linearly independent. Thus $I = I_0$ is finite.

Now let $x_1, \dots, x_r$ and $y_1, \dots, y_s$ be two bases of M . Let $J \subset R$ be a maximal ideal and set $F := R/J$, a field. Then M/JM is naturally an F -vector space. Write $\overline{m}$ for the class of $m \in M$ in M/JM , and $\overline{a}$ for the class of $a \in R$ in F . We claim that $\overline{x_1}, \dots, \overline{x_r}$ form an F -basis of M/JM .

Spanning. For any $m \in M$, write $m = \sum_{i=1}^r a_i x_i$. Then

$$\overline{m} = \sum_{i=1}^r \overline{a_i} \overline{x_i},$$

so $\overline{x_1}, \dots, \overline{x_r}$ span M/JM .

Linear independence. Suppose

$$\sum_{i=1}^r \overline{a_i} \overline{x_i} = 0 \quad \text{in } M/JM.$$

Then $\sum_{i=1}^r a_i x_i \in JM$, so there exist $a'_1, \dots, a'_r \in J$ such that

$$\sum_{i=1}^r a_i x_i = \sum_{i=1}^r a'_i x_i.$$

Hence

$$\sum_{i=1}^r (a_i - a'_i) x_i = 0.$$

By linear independence of $x_1, \dots, x_r$, we get $a_i - a'_i = 0$ for all i , so $a_i \in J$, thus $\overline{a_i} = 0$ for

all i . Therefore $\overline{x_1}, \dots, \overline{x_r}$ are F -linearly independent, and form an F -basis of M/JM .

By the same argument, $\overline{y_1}, \dots, \overline{y_s}$ also form an F -basis of M/JM . Hence

$$r = \dim_F(M/JM) = s.$$

Therefore any two bases of M have the same cardinality. □

Chapter 2

Lecture 2: Tensor Product

2.1 Annihilator, Torsion and Projectivity of Free Modules

Definition 2.1.1. Let M be an R -module, $m \in M$, $Rm = \langle m \rangle = \{rm \mid r \in R\}$. Define the annihilator of m in R as:

$$\text{Ann}_R(m) = \{r \in R \mid rm = 0\}.$$

It is a left ideal of R .

Then we have $R/\text{Ann}_R(m) \cong Rm$ is an isomorphism. Which shows that all cyclic modules are isomorphic to a quotient of R .

Definition 2.1.2. An element $m \in M$ is called

- **torsion-free** if $\text{Ann}_R(m) = 0$.
- **torsion** if $\text{Ann}_R(m) \neq 0$.

A module is called torsion-free if all its elements except of zero are torsion-free.

Definition 2.1.3. The torsion part of M is defined as:

$$\text{Tor}(M) = \{m \in M \mid \text{Ann}_R(m) \neq 0\}.$$

Lemma 2.1.4. If R is an integral domain, then the torsion submodule of M is a submodule of M .

Proof. We verify the submodule criteria.

Non-empty. Since $r \cdot 0_M = 0_M$ for every $r \in R$ and $R \neq 0$, we have $\text{Ann}_R(0_M) = R \neq 0$, so $0_M \in \text{Tor}(M)$.

Closed under addition. Let $m, n \in \text{Tor}(M)$. Then there exist $\lambda, \mu \in R \setminus \{0\}$ such that

$\lambda m = 0$ and $\mu n = 0$. Hence

$$(\lambda\mu)(m+n) = \mu(\lambda m) + \lambda(\mu n) = 0.$$

Since R is an integral domain and $\lambda, \mu \neq 0$, we have $\lambda\mu \neq 0$, so $\text{Ann}_R(m+n) \neq 0$ and $m+n \in \text{Tor}(M)$.

Closed under scalar multiplication. Let $m \in \text{Tor}(M)$ and $r \in R$. Choose $\lambda \in R \setminus \{0\}$ with $\lambda m = 0$. Then

$$\lambda(rm) = r(\lambda m) = 0,$$

so $\text{Ann}_R(rm) \neq 0$ and $rm \in \text{Tor}(M)$.

Therefore $\text{Tor}(M)$ is a submodule of M . □

Example 2.1.5. (1) $0 \in M$ is torsion.

(2) Let R be a commutative ring with 1. Then R is an integral domain if and only if ${}_R R$ is torsion-free. In this case $\text{Tor}({}_R R) = 0$, $\text{Tor}(R^n) = 0$ for all $n \geq 1$.

(3) If I is a nonzero ideal of R , then $\text{Tor}(R/I) = R/I$ where R/I is regarded as an R -module. If we regard R/I as an R/I -module, then it is torsion-free if and only if I is a prime ideal.

(4) More generally, let I be any non-zero ideal of R , and M be an R -module. Then $\text{Tor}(M/IM) = M/IM$ where M/IM is regarded as an R -module.

Remark 2.1.6. Let R be a commutative ring with 1. Let M be an R -module. Then $f : M \rightarrow M$ such that $f(m) = rm$ for all $m \in M$ and $r \in R$ is a homomorphism of R -modules. If moreover M is torsion-free, then f is a monomorphism.

Lemma 2.1.7. If R is an integral domain and M is an R -module, then

1. $\text{Tor}(\text{Tor}(M)) = \text{Tor}(M)$.
2. $\text{Tor}(M/\text{Tor}(M)) = 0$.

Proof.

1. Since $\text{Tor}(M)$ is a submodule of M , its torsion submodule is

$$\text{Tor}(\text{Tor}(M)) = \{ m \in \text{Tor}(M) \mid \text{Ann}_R(m) \neq 0 \}.$$

Every $m \in \text{Tor}(M)$ already satisfies $\text{Ann}_R(m) \neq 0$, so $\text{Tor}(\text{Tor}(M)) \subseteq \text{Tor}(M)$. Conversely, $\text{Tor}(M) \subseteq \text{Tor}(\text{Tor}(M))$ because the annihilator of m in R does not depend on the ambient module: for any $m \in \text{Tor}(M)$, its annihilator computed in the submodule $\text{Tor}(M)$ is the same as $\text{Ann}_R(m) \neq 0$, so $m \in \text{Tor}(\text{Tor}(M))$. Hence $\text{Tor}(\text{Tor}(M)) = \text{Tor}(M)$.

2. Let $\pi : M \rightarrow M/\text{Tor}(M)$ be the canonical projection. Write $\bar{m} = \pi(m) = m +$

$\text{Tor}(M)$. Suppose $\bar{m} \in \text{Tor}(M/\text{Tor}(M))$, i.e. there exists $\lambda \in R \setminus \{0\}$ such that

$$\lambda \bar{m} = \bar{0} \quad \text{in } M/\text{Tor}(M).$$

This means $\lambda m \in \text{Tor}(M)$, so there exists $\mu \in R \setminus \{0\}$ with $\mu(\lambda m) = 0$, i.e.

$$(\mu\lambda)m = 0.$$

Since R is an integral domain and $\mu, \lambda \neq 0$, we have $\mu\lambda \neq 0$, hence $m \in \text{Tor}(M)$, so $\bar{m} = \bar{0}$. Therefore $\text{Tor}(M/\text{Tor}(M)) = 0$.

□

Definition 2.1.8. Let J be a subscript set, denote $R^J = \prod_{j \in J} R$. $R^{(J)} = \bigoplus_{j \in J} R$. Let M be an R -module. Let $x_j \in M$ where $j \in J$ be a family of elements of M . Then the submodule generated by $(x_j)_{j \in J}$ is denoted by $\sum_{j \in J} Rx_j$.

Corollary 2.1.9. We have a morphism of R -modules $\phi : R^{(J)} \rightarrow M$ such that $\phi(r_j)_{j \in J} = \sum_{j \in J} r_j x_j$ for all $(r_j)_{j \in J} \in R^{(J)}$.

- ϕ is injective if and only if $\{x_j\}_{j \in J}$ are linearly independent.
- ϕ is surjective if and only if $\{x_j\}_{j \in J}$ generate M .
- ϕ is an isomorphism if and only if M is free with basis $\{x_j\}_{j \in J}$.

Lemma 2.1.10. Let L be a free R -module with basis $\{x_j\}_{j \in J}$. For any R -module M the following map:

$$\text{Hom}_R(L, M) \rightarrow M^J, \quad f \mapsto (f(x_j))_{j \in J}$$

is a bijection.

Proposition 2.1.11. Let L be a free R -module.

1. Let $\pi : X \rightarrow Y$ be an epimorphism of R -modules. For any R -module homomorphism $\varphi : L \rightarrow Y$ there exists a morphism $\tilde{\varphi} : L \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} & L & \\ \tilde{\varphi} \swarrow & \downarrow \varphi & \\ X & \xrightarrow{\pi} & Y \longrightarrow 0 \end{array}$$

2. In particular, for any R -module M any epimorphism $\pi : M \rightarrow L$ has a section, that is, there exists a morphism $\sigma : L \rightarrow M$ such that $\pi\sigma = \text{id}_L$. In that case σ is injective, $\text{im}(\sigma) = L$ and $M = \ker \pi \oplus \text{im}(\sigma)$.

$$0 \longrightarrow \ker \pi \longrightarrow M \xrightarrow[\sigma]{\varphi} L \longrightarrow 0$$

Proof. Let $\{e_j\}_{j \in J}$ be a basis of L .

(1) Since π is surjective, for each $j \in J$ there exists $\tilde{x}_j \in X$ such that $\pi(\tilde{x}_j) = \varphi(e_j)$. By the lemma above, the map $\text{Hom}_R(L, X) \rightarrow X^J$ given by $f \mapsto (f(e_j))_{j \in J}$ is a bijection. Hence there exists a unique R -module homomorphism $\tilde{\varphi} : L \rightarrow X$ such that $\tilde{\varphi}(e_j) = \tilde{x}_j$ for all $j \in J$. Then for each basis element e_j ,

$$\pi \circ \tilde{\varphi}(e_j) = \pi(\tilde{x}_j) = \varphi(e_j),$$

so $\pi \circ \tilde{\varphi}$ and φ agree on the basis $\{e_j\}_{j \in J}$. Since the basis generates L , we conclude $\pi \circ \tilde{\varphi} = \varphi$.

Remark 2.1.12. *The uniqueness appearing in the proof of the lifting property is not the uniqueness of the lift among all morphisms $\tilde{\varphi} : L \rightarrow X$ satisfying*

$$\pi \circ \tilde{\varphi} = \varphi.$$

Rather, it is only the uniqueness after a choice of lifts of the basis elements has been fixed.

Indeed, let L be a free R -module with basis $\{e_j\}_{j \in J}$. Since $\pi : X \rightarrow Y$ is surjective, for each $j \in J$ we may choose an element $\tilde{x}_j \in X$ such that

$$\pi(\tilde{x}_j) = \varphi(e_j).$$

By the universal property of a free module, or equivalently by the bijection

$$\text{Hom}_R(L, X) \xrightarrow{\sim} X^J, \quad f \mapsto (f(e_j))_{j \in J},$$

there exists a unique R -module homomorphism

$$\tilde{\varphi} : L \rightarrow X$$

such that

$$\tilde{\varphi}(e_j) = \tilde{x}_j \quad \text{for all } j \in J.$$

Thus the word “unique” here means: for the chosen family $(\tilde{x}_j)_{j \in J}$, there is a unique homomorphism realizing those values on the basis.

However, the elements $\tilde{x}_j$ themselves are generally not unique. If $\tilde{x}_j$ is one lift of $\varphi(e_j)$, then for any $k_j \in \ker \pi$, the element

$$\tilde{x}_j + k_j$$

is also a lift, since

$$\pi(\tilde{x}_j + k_j) = \pi(\tilde{x}_j) + \pi(k_j) = \varphi(e_j).$$

Different choices of lifts of the basis elements therefore give, in general, different morphisms $\tilde{\varphi} : L \rightarrow X$. Hence the proposition asserts only existence of a lift, not uniqueness.

Equivalently, if $\tilde{\varphi}_1, \tilde{\varphi}_2 : L \rightarrow X$ are two lifts of φ , then

$$\pi \circ \tilde{\varphi}_1 = \varphi = \pi \circ \tilde{\varphi}_2,$$

so

$$\pi \circ (\tilde{\varphi}_1 - \tilde{\varphi}_2) = 0.$$

Therefore

$$\text{im}(\tilde{\varphi}_1 - \tilde{\varphi}_2) \subseteq \ker \pi,$$

that is,

$$\tilde{\varphi}_1 - \tilde{\varphi}_2 \in \text{Hom}_R(L, \ker \pi).$$

Conversely, if $\delta \in \text{Hom}_R(L, \ker \pi)$, then

$$\pi \circ (\tilde{\varphi}_1 + \delta) = \pi \circ \tilde{\varphi}_1 = \varphi,$$

so $\tilde{\varphi}_1 + \delta$ is again a lift of φ . Thus the lift is unique if and only if

$$\text{Hom}_R(L, \ker \pi) = 0.$$

(2) Apply (1) with $X = M$, $Y = L$, and $\varphi = \text{id}_L$. Then there exists $\sigma : L \rightarrow M$ such that $\pi \circ \sigma = \text{id}_L$.

σ is injective: If $\sigma(l) = 0$, then $l = \text{id}_L(l) = \pi(\sigma(l)) = \pi(0) = 0$.

$M = \ker \pi \oplus \text{im}(\sigma)$: For any $m \in M$, write

$$m = (m - \sigma(\pi(m))) + \sigma(\pi(m)).$$

The second summand $\sigma(\pi(m)) \in \text{im}(\sigma)$. At first, $\pi(m - \sigma(\pi(m))) = \pi(m) - \pi\sigma(\pi(m)) = \pi(m) - \pi(m) = 0$, so it lies in $\ker \pi$. This shows $M = \ker \pi + \text{im}(\sigma)$.

For the directness, if $m \in \ker \pi \cap \text{im}(\sigma)$, write $m = \sigma(l)$ for some $l \in L$. Then $0 = \pi(m) = \pi(\sigma(l)) = l$, so $m = \sigma(0) = 0$. Hence the sum is direct. $\square$

Exercise 2.1.13. Let L be a free R -module. P a direct summand of L . Show that whenever $\pi : X \rightarrow Y$ is an epimorphism and $\varphi : P \rightarrow Y$ is a morphism, there exists a morphism $\tilde{\varphi} : P \rightarrow X$ such that $\varphi = \pi \circ \tilde{\varphi}$.

2.2 Tensor product

Let R be a commutative ring. M, N be R -modules. The tensor product $M \otimes_R N$ is a "linearize" of the bilinear map.

Definition 2.2.1. Let R be a commutative ring. M, N, K be R -modules. $f : M \times N \rightarrow K$ is a bilinear map if for all $r \in R$, $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, we have:

- $f(m_1 + m_2, n) = f(m_1, n) + f(m_2, n)$

- $f(m, n_1 + n_2) = f(m, n_1) + f(m, n_2)$
- $f(rm, n) = rf(m, n)$
- $f(m, rn) = rf(m, n)$

Example 2.2.2. If $f : M \times N \rightarrow K$ is a bilinear map, then $f(x, 0) = 0 = f(0, y)$ for all $x \in M, y \in N$. $f(-x, y) = -f(x, y)$ and $f(x, -y) = -f(x, y)$ for all $x \in M, y \in N$.

Definition 2.2.3. The tensor product of M and N is a pair $(M \otimes_R N, t_{M,N})$ where $M \otimes_R N$ is an R -module and $t_{M,N} : M \times N \rightarrow M \otimes_R N$ is a bilinear map usually denoted by $t_{M,N}(m, n) = m \otimes n$ which satisfies the following universal properties: Whenever X is an R -module and $f : M \times N \rightarrow X$ is a bilinear map, there exists a unique R -module homomorphism $\tilde{f} : M \otimes_R N \rightarrow X$ such that the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{f} & X \\ t_{M,N} \downarrow & \nearrow \tilde{f} & \\ M \otimes_R N & & \end{array}$$

Theorem 2.2.4. The tensor product of M and N exists and is unique up to isomorphism.

Proof. We prove the uniqueness at first.

The tensor product is unique up to isomorphism. Suppose (T_1, t_1) and (T_2, t_2) are two tensor products of M and N . By the universal property, there exists a unique R -module homomorphism $\varphi : T_1 \rightarrow T_2$ such that $t_2 = \varphi \circ t_1$ and a unique R -module homomorphism $\psi : T_2 \rightarrow T_1$ such that $t_1 = \psi \circ t_2$. Then we have:

$$\varphi \circ \psi \circ t_2 = \varphi \circ t_1 = t_2$$

$$\psi \circ \varphi \circ t_1 = \psi \circ t_2 = t_1.$$

Since such $\varphi \circ \psi$ and $\psi \circ \varphi$ are unique. It forces $\varphi \circ \psi = \text{id}_{T_2}$ and $\psi \circ \varphi = \text{id}_{T_1}$. Hence φ and ψ are isomorphisms. Therefore $T_1 \cong T_2$.

Now we prove the existence.

Let F be a free R -module with basis $M \times N$. i.e. $F = \text{span}_R(e_{m_i, n_i})_{m_i \in M, n_i \in N}$. Let B be the submodule of F generated by the elements of the following forms:

- $e_{m_1+m_2, n} - e_{m_1, n} - e_{m_2, n}$
- $e_{m, n_1+n_2} - e_{m, n_1} - e_{m, n_2}$
- $e_{rm, n} - re_{m, n}$

- $e_{m,rn} - re_{m,n}$

Where $m, m_1, m_2 \in M$, $n, n_1, n_2 \in N$, $r \in R$. Define $M \otimes_R N = F/B$, $t_{M,N} : M \times N \rightarrow M \otimes_R N$ by $t_{M,N}(m, n) = e_{m,n} + B$. It is easy to check that $t_{M,N}$ is a well-defined bilinear map.

We only need to show that $(M \otimes_R N, t_{M,N})$ satisfies the universal property of tensor product:

Let X be an R -module and $f : M \times N \rightarrow X$ a bilinear map. Since F is a free R -module with basis $\{e_{m,n}\}_{(m,n) \in M \times N}$, by the universal property of the free module, there exists a unique R -module homomorphism $\hat{f} : F \rightarrow X$ such that $\hat{f}(e_{m,n}) = f(m, n)$ for all $(m, n) \in M \times N$.

We claim that $B \subseteq \ker \hat{f}$. Indeed, since f is bilinear:

$$\begin{aligned}\hat{f}(e_{m_1+m_2,n} - e_{m_1,n} - e_{m_2,n}) &= f(m_1 + m_2, n) - f(m_1, n) - f(m_2, n) = 0, \\ \hat{f}(e_{m,n_1+n_2} - e_{m,n_1} - e_{m,n_2}) &= f(m, n_1 + n_2) - f(m, n_1) - f(m, n_2) = 0, \\ \hat{f}(e_{rm,n} - re_{m,n}) &= f(rm, n) - rf(m, n) = 0, \\ \hat{f}(e_{m,rn} - re_{m,n}) &= f(m, rn) - rf(m, n) = 0.\end{aligned}$$

Since B is generated by elements of the above forms and $\ker \hat{f}$ is a submodule, we have $B \subseteq \ker \hat{f}$.

By the universal property of quotient modules, $\hat{f}$ induces a unique R -module homomorphism $\tilde{f} : M \otimes_R N = F/B \rightarrow X$ such that $\tilde{f} \circ \pi = \hat{f}$, where $\pi : F \rightarrow F/B$ is the canonical projection. In particular, $\tilde{f}(m \otimes n) = \tilde{f}(e_{m,n} + B) = \hat{f}(e_{m,n}) = f(m, n)$, so $\tilde{f} \circ t_{M,N} = f$.

For uniqueness, suppose $g : M \otimes_R N \rightarrow X$ is another R -module homomorphism with $g \circ t_{M,N} = f$. Then $g(m \otimes n) = f(m, n) = \tilde{f}(m \otimes n)$ for all $m \in M, n \in N$. Since elements of the form $m \otimes n$ generate $M \otimes_R N$ as an R -module, $g = \tilde{f}$.

Therefore $(M \otimes_R N, t_{M,N})$ is a tensor product of M and N . □

Definition 2.2.5. The set of elements of the form $m \otimes n$ for $m \in M$ and $n \in N$ is called the set of elementary tensors. This set generates $M \otimes_R N$ as an R -module.

Example 2.2.6. Fix $m \in M$. The map $N \rightarrow M \otimes_R N, n \mapsto m \otimes n$ is a morphism of R -modules. Its image $m \otimes_R N = \{m \otimes n \mid n \in N\}$ is a submodule of $M \otimes_R N$ which is isomorphic to a quotient $N / \text{Ann}_R(m)N$.

Proposition 2.2.7. We have the following isomorphisms:

1. $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$ where $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$.
2. $M_1 \otimes_R M_2 \cong M_2 \otimes_R M_1$ where $m_1 \otimes m_2 \mapsto m_2 \otimes m_1$.
3. $M_1 \otimes_R (M_2 \oplus M_3) \cong (M_1 \otimes_R M_2) \oplus (M_1 \otimes_R M_3)$ where $m_1 \otimes (m_2 \oplus m_3) \mapsto$

$(m_1 \otimes m_2) + (m_1 \otimes m_3)$. And more generally,

$$M_1 \otimes_R \left(\bigoplus_{i=1}^n M_i \right) \cong \bigoplus_{i=1}^n (M_1 \otimes_R M_i)$$

where $m_1 \otimes (m_2 \oplus \cdots \oplus m_n) \mapsto (m_1 \otimes m_2) + \cdots + (m_1 \otimes m_n)$.

4. $R \otimes_R M \cong M$ where $r \otimes m \mapsto rm$. And more generally, $R^{(I)} \otimes_R M \cong M^{(I)}$ where $(r_i)_{i \in I} \otimes m \mapsto (r_i m)_{i \in I}$.

Proof. We prove (1): $(M_1 \otimes_R M_2) \otimes_R M_3 \cong M_1 \otimes_R (M_2 \otimes_R M_3)$.

Step 1. We construct $\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$.

Fix $m_3 \in M_3$. For each m_3 , define $f_{m_3} : M_1 \times M_2 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ by $f_{m_3}(m_1, m_2) = m_1 \otimes (m_2 \otimes m_3)$. This is bilinear in (m_1, m_2) , so by the universal property of $M_1 \otimes_R M_2$, there exists a unique R -module homomorphism $\tilde{f}_{m_3} : M_1 \otimes_R M_2 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ such that $\tilde{f}_{m_3}(m_1 \otimes m_2) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{array}{ccc} M_1 \times M_2 & \xrightarrow{f_{m_3}} & M_1 \otimes_R (M_2 \otimes_R M_3) \\ \downarrow t_{M_1, M_2} & \nearrow \tilde{f}_{m_3} & \\ M_1 \otimes_R M_2 & & \end{array}$$

Now define $g : (M_1 \otimes_R M_2) \times M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$ by $g(\alpha, m_3) = \tilde{f}_{m_3}(\alpha)$. We verify that g is bilinear. Linearity in the first argument follows because each $\tilde{f}_{m_3}$ is an R -module homomorphism. For the second argument, on elementary tensors:

$$\begin{aligned} g(m_1 \otimes m_2, m_3 + m'_3) &= m_1 \otimes (m_2 \otimes (m_3 + m'_3)) \\ &= m_1 \otimes (m_2 \otimes m_3 + m_2 \otimes m'_3) \\ &= m_1 \otimes (m_2 \otimes m_3) + m_1 \otimes (m_2 \otimes m'_3) \\ &= g(m_1 \otimes m_2, m_3) + g(m_1 \otimes m_2, m'_3), \\ g(m_1 \otimes m_2, rm_3) &= m_1 \otimes (m_2 \otimes rm_3) \\ &= m_1 \otimes (r(m_2 \otimes m_3)) \\ &= r(m_1 \otimes (m_2 \otimes m_3)) \\ &= r g(m_1 \otimes m_2, m_3). \end{aligned}$$

Since elementary tensors generate $M_1 \otimes_R M_2$, bilinearity holds on all of $(M_1 \otimes_R M_2) \times M_3$.

By the universal property of $(M_1 \otimes_R M_2) \otimes_R M_3$, there exists a unique R -module homomorphism

$$\varphi : (M_1 \otimes_R M_2) \otimes_R M_3 \rightarrow M_1 \otimes_R (M_2 \otimes_R M_3)$$

such that $\varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3)$:

$$\begin{array}{ccc} (M_1 \otimes_R M_2) \times M_3 & \xrightarrow{g} & M_1 \otimes_R (M_2 \otimes_R M_3) \\ \downarrow t & \nearrow \varphi & \\ (M_1 \otimes_R M_2) \otimes_R M_3 & & \end{array}$$

Step 2. By an entirely symmetric argument (fixing m_1 first, then extending), we obtain an R -module homomorphism

$$\psi : M_1 \otimes_R (M_2 \otimes_R M_3) \rightarrow (M_1 \otimes_R M_2) \otimes_R M_3$$

such that $\psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3$.

Step 3. We show φ and ψ are inverses. On elementary tensors:

$$(\psi \circ \varphi)((m_1 \otimes m_2) \otimes m_3) = \psi(m_1 \otimes (m_2 \otimes m_3)) = (m_1 \otimes m_2) \otimes m_3,$$

$$(\varphi \circ \psi)(m_1 \otimes (m_2 \otimes m_3)) = \varphi((m_1 \otimes m_2) \otimes m_3) = m_1 \otimes (m_2 \otimes m_3).$$

Since elementary tensors generate both modules, $\psi \circ \varphi = \text{id}$ and $\varphi \circ \psi = \text{id}$. The overall situation is summarized by the following commutative diagram:

$$\begin{array}{ccc} & M_1 \times M_2 \times M_3 & \\ \swarrow & & \searrow \\ (M_1 \otimes_R M_2) \otimes_R M_3 & \xrightleftharpoons[\psi]{\varphi} & M_1 \otimes_R (M_2 \otimes_R M_3) \end{array}$$

where the diagonal arrows send (m_1, m_2, m_3) to $(m_1 \otimes m_2) \otimes m_3$ and $m_1 \otimes (m_2 \otimes m_3)$ respectively. Hence φ is an isomorphism with $(m_1 \otimes m_2) \otimes m_3 \mapsto m_1 \otimes (m_2 \otimes m_3)$. $\square$

Theorem 2.2.8. Let M, N, P be R -modules. We have a canonical isomorphism:

$$\phi_{M,N,P} : \text{Hom}_R(M \otimes_R N, P) \rightarrow \text{Hom}_R(M, \text{Hom}_R(N, P))$$

such that $\phi_{M,N,P}(f)(m)(n) = f(m \otimes n)$. Furthermore, if R is commutative, then $\phi_{M,N,P}$ is an R -module isomorphism. If R is not commutative, then $\phi_{M,N,P}$ is only a $\mathbb{Z}$ -module isomorphism.

Proof. Let $f \in \text{Hom}_R(M \otimes_R N, P)$. For each $m \in M$, define $\phi_{M,N,P}(f)(m) : N \rightarrow P$ by

$$\phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

We verify that $\phi_{M,N,P}(f)(m) \in \text{Hom}_R(N, P)$: for $n_1, n_2 \in N$ and $r \in R$,

$$\begin{aligned} \phi_{M,N,P}(f)(m)(n_1 + n_2) &= f(m \otimes (n_1 + n_2)) = f(m \otimes n_1 + m \otimes n_2) \\ &= f(m \otimes n_1) + f(m \otimes n_2), \\ \phi_{M,N,P}(f)(m)(rn) &= f(m \otimes rn) = f(r(m \otimes n)) \\ &= rf(m \otimes n) = r\phi_{M,N,P}(f)(m)(n). \end{aligned}$$

Next, we verify that $\phi_{M,N,P}(f) \in \text{Hom}_R(M, \text{Hom}_R(N, P))$: for $m_1, m_2 \in M$, $r \in R$, and

any $n \in N$,

$$\begin{aligned}
\phi_{M,N,P}(f)(m_1 + m_2)(n) &= f((m_1 + m_2) \otimes n) \\
&= f(m_1 \otimes n) + f(m_2 \otimes n) \\
\phi_{M,N,P}(f)(rm)(n) &= f(rm \otimes n) \\
&= f(r(m \otimes n)) \\
&= rf(m \otimes n) \\
&= (r \phi_{M,N,P}(f)(m))(n).
\end{aligned}$$

So

$$\phi_{M,N,P}(f)(m_1 + m_2) = \phi_{M,N,P}(f)(m_1) + \phi_{M,N,P}(f)(m_2)$$

and $\phi_{M,N,P}(f)(rm) = r \phi_{M,N,P}(f)(m)$, confirming R -linearity.

Let $g \in \text{Hom}_R(M, \text{Hom}_R(N, P))$. Define $\hat{g} : M \times N \rightarrow P$ by $\hat{g}(m, n) = g(m)(n)$. We check that $\hat{g}$ is R -bilinear:

$$\begin{aligned}
\hat{g}(m_1 + m_2, n) &= g(m_1 + m_2)(n) = g(m_1)(n) + g(m_2)(n) = \hat{g}(m_1, n) + \hat{g}(m_2, n), \\
\hat{g}(m, n_1 + n_2) &= g(m)(n_1 + n_2) = g(m)(n_1) + g(m)(n_2) = \hat{g}(m, n_1) + \hat{g}(m, n_2), \\
\hat{g}(rm, n) &= g(rm)(n) = (rg(m))(n) = rg(m)(n) = r \hat{g}(m, n), \\
\hat{g}(m, rn) &= g(m)(rn) = rg(m)(n) = r \hat{g}(m, n).
\end{aligned}$$

By the universal property of $M \otimes_R N$, there exists a unique R -module homomorphism $\psi_{M,N,P}(g) : M \otimes_R N \rightarrow P$ such that $\psi_{M,N,P}(g)(m \otimes n) = \hat{g}(m, n)$:

$$\begin{array}{ccc}
M \times N & \xrightarrow{\hat{g}} & P \\
t_{M,N} \downarrow & \nearrow \psi_{M,N,P}(g) & \\
M \otimes_R N & &
\end{array}$$

For any $f \in \text{Hom}_R(M \otimes_R N, P)$:

$$\psi_{M,N,P}(\phi_{M,N,P}(f))(m \otimes n) = \phi_{M,N,P}(f)(m)(n) = f(m \otimes n).$$

Since elementary tensors generate $M \otimes_R N$, we have $\psi_{M,N,P}(\phi_{M,N,P}(f)) = f$.

For any $g \in \text{Hom}_R(M, \text{Hom}_R(N, P))$, for all $m \in M$ and $n \in N$:

$$\phi_{M,N,P}(\psi_{M,N,P}(g))(m)(n) = \psi_{M,N,P}(g)(m \otimes n) = g(m)(n).$$

So $\phi_{M,N,P}(\psi_{M,N,P}(g))(m) = g(m)$ for all m , hence $\phi_{M,N,P}(\psi_{M,N,P}(g)) = g$.

Therefore $\phi_{M,N,P}$ is a bijection of sets (in fact, of abelian groups).

It remains to show that $\phi_{M,N,P}$ respects the appropriate module structure.

Both $\text{Hom}_R(M \otimes_R N, P)$ and $\text{Hom}_R(M, \text{Hom}_R(N, P))$ are abelian groups under point-

wise addition. We verify $\phi_{M,N,P}$ is a group homomorphism: for $f_1, f_2 \in \text{Hom}_R(M \otimes_R N, P)$,

$$\begin{aligned}\phi_{M,N,P}(f_1 + f_2)(m)(n) &= (f_1 + f_2)(m \otimes n) \\ &= f_1(m \otimes n) + f_2(m \otimes n) \\ &= \phi_{M,N,P}(f_1)(m)(n) + \phi_{M,N,P}(f_2)(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(f_1 + f_2) = \phi_{M,N,P}(f_1) + \phi_{M,N,P}(f_2)$. Hence $\phi_{M,N,P}$ is a $\mathbb{Z}$ -module isomorphism.

If R is commutative, $\text{Hom}_R(M \otimes_R N, P)$ carries an R -module structure via $(r \cdot f)(x) = rf(x)$, and similarly for $\text{Hom}_R(M, \text{Hom}_R(N, P))$. Then

$$\begin{aligned}\phi_{M,N,P}(rf)(m)(n) &= (rf)(m \otimes n) \\ &= rf(m \otimes n) \\ &= r \phi_{M,N,P}(f)(m)(n) \\ &= (r \phi_{M,N,P}(f))(m)(n)\end{aligned}$$

so $\phi_{M,N,P}(rf) = r \phi_{M,N,P}(f)$, and $\phi_{M,N,P}$ is an R -module isomorphism.

If R is not commutative, $\text{Hom}_R(M \otimes_R N, P)$ does not naturally carry an R -module structure (scalar multiplication by r need not commute with R -linearity), so $\phi_{M,N,P}$ is only a $\mathbb{Z}$ -module isomorphism. $\square$

Definition 2.2.9. Let $f : M \rightarrow M'$ and $g : N \rightarrow N'$ be R -module homomorphisms. The tensor product of f and g is the R -module homomorphism:

$$f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$$

defined by $(f \otimes g)(m \otimes n) = f(m) \otimes g(n)$.

Definition 2.2.10. Let $M_i, i \in I$ be a family of R -modules. Use the following notation:

$$L^n(M_1, \dots, M_n; N) = \{f : M_1 \times \dots \times M_n \rightarrow N \mid f \text{ is } n\text{-linear}\}$$

In particular,

$$L^2(M_1, M_2; N) \cong L(M_1 \otimes M_2, N) \cong \text{Hom}_R(M_1 \otimes M_2, N)$$

or equivalently,

$$L^2(M_1 \otimes M_2, N) \cong L(M_1, L(M_2, N))$$

Proposition 2.2.11. Let M, N be two finitely generated free R -modules. Denote

$$M^\vee = \text{Hom}_R(M, R)$$

Then we have a canonical isomorphism:

$$M^\vee \otimes N \cong \text{Hom}_R(M, N)$$

Proof. Since M is a finitely generated free R -module, we have $M \cong R^m$ for some $m \geq 1$. We construct an explicit isomorphism

$$\Phi : M^\vee \otimes_R N \rightarrow \text{Hom}_R(M, N).$$

Define a bilinear map $\beta : M^\vee \times N \rightarrow \text{Hom}_R(M, N)$ by

$$\beta(f, n)(m) = f(m) n, \quad f \in M^\vee, n \in N, m \in M.$$

We check R -bilinearity. For $f_1, f_2 \in M^\vee$, $n, n_1, n_2 \in N$, $m \in M$, and $r \in R$:

$$\begin{aligned} \beta(f_1 + f_2, n)(m) &= (f_1 + f_2)(m) n = f_1(m) n + f_2(m) n \\ &= \beta(f_1, n)(m) + \beta(f_2, n)(m), \\ \beta(f, n_1 + n_2)(m) &= f(m)(n_1 + n_2) = f(m) n_1 + f(m) n_2 \\ &= \beta(f, n_1)(m) + \beta(f, n_2)(m), \\ \beta(rf, n)(m) &= (rf)(m) n = r f(m) n = r \beta(f, n)(m), \\ \beta(f, rn)(m) &= f(m) (rn) = r f(m) n = r \beta(f, n)(m). \end{aligned}$$

By the universal property of the tensor product, there exists a unique R -module homomorphism

$$\Phi : M^\vee \otimes_R N \rightarrow \text{Hom}_R(M, N), \quad \Phi(f \otimes n)(m) = f(m) n.$$

Now let $\{e_1, \dots, e_m\}$ be a basis of $M \cong R^m$, and let $\{e_1^*, \dots, e_m^*\}$ be the dual basis, i.e. $e_i^* \in M^\vee$ with $e_i^*(e_j) = \delta_{ij}$. Define

$$\Psi : \text{Hom}_R(M, N) \rightarrow M^\vee \otimes_R N, \quad \Psi(g) = \sum_{i=1}^m e_i^* \otimes g(e_i).$$

We verify Ψ is an R -module homomorphism: for $g_1, g_2 \in \text{Hom}_R(M, N)$ and $r \in R$,

$$\begin{aligned} \Psi(g_1 + g_2) &= \sum_{i=1}^m e_i^* \otimes (g_1 + g_2)(e_i) = \sum_{i=1}^m e_i^* \otimes (g_1(e_i) + g_2(e_i)) \\ &= \sum_{i=1}^m e_i^* \otimes g_1(e_i) + \sum_{i=1}^m e_i^* \otimes g_2(e_i) = \Psi(g_1) + \Psi(g_2), \\ \Psi(rg) &= \sum_{i=1}^m e_i^* \otimes (rg)(e_i) = \sum_{i=1}^m e_i^* \otimes r g(e_i) \\ &= r \sum_{i=1}^m e_i^* \otimes g(e_i) = r \Psi(g). \end{aligned}$$

We show Φ and Ψ are mutual inverses. For any $g \in \text{Hom}_R(M, N)$ and $m = \sum_j r_j e_j \in M$:

$$\begin{aligned}\Phi(\Psi(g))(m) &= \Phi\left(\sum_{i=1}^m e_i^* \otimes g(e_i)\right)\left(\sum_j r_j e_j\right) \\ &= \sum_{i=1}^m e_i^*\left(\sum_j r_j e_j\right)g(e_i) = \sum_{i=1}^m r_i g(e_i) \\ &= g\left(\sum_{i=1}^m r_i e_i\right) = g(m).\end{aligned}$$

So $\Phi \circ \Psi = \text{id}$.

For any elementary tensor $f \otimes n \in M^\vee \otimes_R N$:

$$\begin{aligned}\Psi(\Phi(f \otimes n)) &= \sum_{i=1}^m e_i^* \otimes \Phi(f \otimes n)(e_i) = \sum_{i=1}^m e_i^* \otimes f(e_i) n \\ &= \sum_{i=1}^m f(e_i) (e_i^* \otimes n) = \left(\sum_{i=1}^m f(e_i) e_i^*\right) \otimes n = f \otimes n,\end{aligned}$$

where the last equality uses $f = \sum_{i=1}^m f(e_i) e_i^*$ (expansion of f in the dual basis). Since elementary tensors generate $M^\vee \otimes_R N$ and $\Psi \circ \Phi$ is R -linear, we have $\Psi \circ \Phi = \text{id}$.

Therefore Φ is an R -module isomorphism, giving $M^\vee \otimes_R N \cong \text{Hom}_R(M, N)$. $\square$

Proposition 2.2.12. *Assume that R is a subring of S , and M is an R -module. Then the bilinear map:*

$$S \times (S \otimes_R M) \rightarrow S \otimes_R M$$

defined by

$$(s, s' \otimes m) \mapsto ss' \otimes m$$

defines a structure of S -module on the R -module $S \otimes_R M$.

More generally, let $f : R \rightarrow T$ be a ring homomorphism. And view T as an R -module via $R \times T \rightarrow T, (\lambda, \mu) \mapsto f(\lambda)\mu$. Then the bilinear map:

$$T \times (T \otimes_R M) \rightarrow T \otimes_R M$$

defined by

$$(t, t' \otimes m) \mapsto tt' \otimes m$$

defines a structure of T -module on the R -module $T \otimes_R M$.

Proposition 2.2.13. *Let I be an ideal of R . Let M be an R -module. Then the morphism of R -modules:*

$$M \rightarrow (R/I) \otimes_R M$$

defined by

$$m \mapsto 1_{R/I} \otimes m$$

induces an isomorphism of R -modules:

$$M/(IM) \cong (R/I) \otimes_R M$$

as R/I -modules.

2.3 Balanced product

Definition 2.3.1. Let R be a ring (not necessarily commutative). Let M_R be a right R -module and ${}_R N$ be a left R -module. The balanced product of M and N is defined as a pair (P, f) where P is an abelian group and $f : M \times N \rightarrow P$ is a bilinear map satisfies:

- $f(rm, n) = f(m, rn)$ for all $r \in R, m \in M, n \in N$.
- $f(m + m', n) = f(m, n) + f(m', n)$ for all $m, m' \in M, n \in N$.
- $f(m, n + n') = f(m, n) + f(m, n')$ for all $m \in M, n, n' \in N$.

Definition 2.3.2. A tensor product of M_R and ${}_R N$ is a balanced product $(M \otimes_R N, t_{M,N})$ such that if (P, f) is any balanced product of M_R and ${}_R N$, then there exists a unique abelian group homomorphism $\tilde{f} : M \otimes_R N \rightarrow P$ such that the following diagram commutes:

$$\begin{array}{ccc} M \times N & \xrightarrow{t_{M,N}} & M \otimes_R N \\ f \downarrow & \tilde{f} \swarrow & \\ P & & \end{array}$$

Chapter 3

Lecture 3: Categories, Bimodules

3.1 Categories

Definition 3.1.1. A category $\mathcal{C}$ consists of:

- A class of objects $\text{ob}(\mathcal{C})$.
- A set of morphisms $\text{Hom}_{\mathcal{C}}(A, B)$ for each pair of objects $A, B \in \mathcal{C}$.
- A composition map $\text{Hom}_{\mathcal{C}}(B, C) \times \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{C}}(A, C)$ such that $(\phi, \psi) \mapsto \phi \circ \psi$ for each triple of objects $A, B, C \in \mathcal{C}$.

It is assumed that objects and morphisms satisfy the following conditions:

- If $(A, B) \neq (C, D)$, then

$$\text{Hom}(A, B) \cap \text{Hom}(C, D) = \emptyset.$$

- If

$$f \in \text{Hom}(A, B), \quad g \in \text{Hom}(B, C), \quad h \in \text{Hom}(C, D),$$

then

$$(h \circ g) \circ f = h \circ (g \circ f).$$

- (Unit) For every object A we have an element

$$1_A \in \text{Hom}(A, A)$$

such that

$$f \circ 1_A = f \quad \text{for any } f \in \text{Hom}(A, B),$$

and

$$1_A \circ g = g \quad \text{for any } g \in \text{Hom}(B, A).$$

Example 3.1.2. *The following are examples of categories.*

1. *The category of sets: the objects are sets, and the morphisms are maps. It is denoted by **Set**.*
2. *The category of groups: the objects are groups, and the morphisms are group homomorphisms. It is denoted by **Grp**.*
3. *The category of abelian groups: the objects are abelian groups, and the morphisms are group homomorphisms. It is denoted by **Ab**.*
4. *The category of R -modules: the objects are R -modules, and the morphisms are R -module homomorphisms. It is denoted by $R\text{-}\mathbf{Mod}$ (left R -modules, for right R -modules, we use $R\text{-}\mathbf{Mod}$).*
5. *The category of finitely generated R -modules: the objects are finitely generated R -modules, and the morphisms are R -module homomorphisms. It is denoted by $R\text{-}\mathbf{mod}$.*
6. *The category of topological spaces: the objects are topological spaces, and the morphisms are continuous maps. It is denoted by **Top**.*

Definition 3.1.3 (Subcategory). *A category $\mathcal{D}$ is called a subcategory of a category $\mathcal{C}$ if*

1. *the objects of $\mathcal{D}$ form a subclass of the objects of $\mathcal{C}$;*
2. *for any objects A, B of $\mathcal{D}$, we have*
 - $\mathrm{Hom}_{\mathcal{D}}(A, B) \subseteq \mathrm{Hom}_{\mathcal{C}}(A, B)$;
 - *the identity morphisms and the composition of morphisms in $\mathcal{D}$ are the same as those in $\mathcal{C}$.*

The subcategory $\mathcal{D}$ of $\mathcal{C}$ is called full if

$$\mathrm{Hom}_{\mathcal{D}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(A, B)$$

for any objects A, B of $\mathcal{D}$.

Example 3.1.4. *The category **Ab** is a full subcategory of **Grp**.*

The category $R\text{-}\mathbf{mod}$ is a full subcategory of $R\text{-}\mathbf{Mod}$.

*The category **Grp** is not a full subcategory of **Set**.*

Definition 3.1.5. *Let $\mathcal{C}$ and $\mathcal{D}$ be categories. A covariant functor*

$$F : \mathcal{C} \rightarrow \mathcal{D}$$

consists of the following data:

- for every object A of $\mathcal{C}$, an object $F(A)$ of $\mathcal{D}$;
- for each pair of objects A, B of $\mathcal{C}$, a map

$$F : \text{Hom}_{\mathcal{C}}(A, B) \rightarrow \text{Hom}_{\mathcal{D}}(F(A), F(B)), \quad f \mapsto F(f).$$

These data are required to satisfy the following conditions:

1. If $g \circ f$ is defined in $\mathcal{C}$, then

$$F(g) \circ F(f) = F(g \circ f).$$

2. For every object A of $\mathcal{C}$,

$$F(1_A) = 1_{F(A)}.$$

Proposition 3.1.6. *Functors preserve diagram:*

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow g & \downarrow h \\ & & C \end{array} \quad \xrightarrow{F} \quad \begin{array}{ccc} F(A) & \xrightarrow{F(f)} & F(B) \\ & \searrow F(g) & \downarrow F(h) \\ & & F(C) \end{array}$$

Let M and N be R -modules. Then

$$\text{Hom}_R(M, N) = \{ f : M \rightarrow N \mid f \text{ is an } R\text{-module homomorphism} \}.$$

This is an abelian group.

Moreover,

$$\text{End}_R(M) = \text{Hom}_R(M, M),$$

and $\text{End}_R(M)$ is a ring.

Example 3.1.7. *Let M be an R -module. Then*

$$\text{Hom}_R(M, -) : R\text{-Mod} \rightarrow \text{Ab}$$

is a functor.

More precisely, it acts as follows:

$$N \longmapsto \text{Hom}_R(M, N), \quad f : N \rightarrow N' \longmapsto \text{Hom}_R(M, f).$$

Given an R -module homomorphism

$$f : N \rightarrow N',$$

the induced map

$$\mathrm{Hom}_R(M, f) : \mathrm{Hom}_R(M, N) \rightarrow \mathrm{Hom}_R(M, N')$$

is defined by

$$\mathrm{Hom}_R(M, f)(h) = f \circ h \quad (h \in \mathrm{Hom}_R(M, N)).$$

This corresponds to the commutative diagram

$$\begin{array}{ccc} & M & \\ h \swarrow & & \searrow f \circ h \\ N & \xrightarrow{f} & N' \end{array}$$

Hence

$$\mathrm{Hom}_R(M, -)$$

is a covariant functor.

Definition 3.1.8. Let $\mathcal{C}$ be a category. The opposite category (or dual category) $\mathcal{C}^{\mathrm{op}}$ is defined as follows:

- $\mathrm{Ob}(\mathcal{C}^{\mathrm{op}}) = \mathrm{Ob}(\mathcal{C})$;
- for any objects A, B ,

$$\mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B) = \mathrm{Hom}_{\mathcal{C}}(B, A).$$

The composition in $\mathcal{C}^{\mathrm{op}}$ is defined by reversing the order of composition in $\mathcal{C}$: if

$$f \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(A, B), \quad g \in \mathrm{Hom}_{\mathcal{C}^{\mathrm{op}}}(B, C),$$

then

$$g \circ f \text{ in } \mathcal{C}^{\mathrm{op}}$$

is defined to be

$$f \circ g \text{ in } \mathcal{C}.$$

We call $\mathcal{C}^{\mathrm{op}}$ the dual category of $\mathcal{C}$.

This is illustrated by the following diagrams:

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ & \searrow g \circ f & \downarrow g \\ & & C \end{array} \quad \text{in } \mathcal{C}^{\mathrm{op}},$$

while the corresponding picture in $\mathcal{C}$ is

$$\begin{array}{ccc} A & \xleftarrow{f} & B \\ & \nwarrow f \circ g & \uparrow g \\ & & C \end{array} \quad \text{in } \mathcal{C}.$$

Definition 3.1.9. A contravariant functor from a category $\mathcal{C}$ to a category $\mathcal{D}$ is a covariant functor

$$\mathcal{C}^{\text{op}} \rightarrow \mathcal{D}.$$

Example 3.1.10. Let M be an R -module. Then

$$\text{Hom}_R(-, M) : R\text{-Mod} \rightarrow \text{Ab}$$

is a contravariant functor.

More precisely, it is defined on objects by

$$N \longmapsto \text{Hom}_R(N, M),$$

and on morphisms by sending an R -module homomorphism

$$f : N \rightarrow N'$$

to the induced group homomorphism

$$\text{Hom}_R(f, M) : \text{Hom}_R(N', M) \rightarrow \text{Hom}_R(N, M),$$

given by

$$\text{Hom}_R(f, M)(h) = h \circ f \quad (h \in \text{Hom}_R(N', M)).$$

This corresponds to the commutative diagram

$$\begin{array}{ccc} N & & \\ f \downarrow & \searrow h \circ f & \\ N' & \xrightarrow{h} & M \end{array}$$

3.2 Exact Sequences

Definition 3.2.1. Let $\{M_i\}_{i \in \mathbb{Z}}$ be a family of R -modules, and let

$$\varphi_i : M_i \rightarrow M_{i+1}$$

be R -module homomorphisms for each $i \in \mathbb{Z}$. Then

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is called a sequence of R -modules.

Definition 3.2.2. The sequence

$$\cdots \longrightarrow M_{i-1} \xrightarrow{\varphi_{i-1}} M_i \xrightarrow{\varphi_i} M_{i+1} \longrightarrow \cdots$$

is said to be exact at M_i if

$$\text{Im}(\varphi_{i-1}) = \ker(\varphi_i).$$

It is called an exact sequence if it is exact at every term.

A short exact sequence is a sequence of the form

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0.$$

It is exact if and only if

$$\ker(\iota) = 0, \quad \text{Im}(\pi) = M'', \quad \text{Im}(\iota) = \ker(\pi).$$

Equivalently, ι is injective, π is surjective, and

$$\text{Im}(\iota) = \ker(\pi).$$

Hence, for every short exact sequence

$$0 \longrightarrow M' \xrightarrow{\iota} M \xrightarrow{\pi} M'' \longrightarrow 0,$$

we have

$$M/M' \cong M''.$$

Definition 3.2.3. Let R, S be rings, and let

$$F : R\text{-Mod} \rightarrow S\text{-Mod}$$

be a covariant functor.

We say that F is left exact if for any exact sequence

$$0 \rightarrow N \xrightarrow{\varphi} M \xrightarrow{\psi} P,$$

the sequence

$$0 \rightarrow F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P)$$

is exact.

We say that F is right exact if for any exact sequence

$$N \xrightarrow{\varphi} M \xrightarrow{\psi} P \rightarrow 0,$$

the sequence

$$F(N) \xrightarrow{F(\varphi)} F(M) \xrightarrow{F(\psi)} F(P) \rightarrow 0$$

is exact.

Definition 3.2.4. Let

$$G : R\text{-Mod} \rightarrow S\text{-Mod}$$

be a contravariant functor. We say that G is left exact (respectively, right exact) if it is left exact (respectively, right exact) when regarded as a covariant functor

$$G : (R\text{-Mod})^{\text{op}} \rightarrow S\text{-Mod}.$$

A functor is called exact if it is both left exact and right exact.

Let M be an R -module. Then:

$$\text{Hom}_R(M, -) : R\text{-Mod} \rightarrow \text{Ab} \quad \text{is left exact,}$$

$$\text{Hom}_R(-, M) : R\text{-Mod} \rightarrow \text{Ab} \quad \text{is right exact,}$$

and

$$M \otimes_R - : R\text{-Mod} \rightarrow \text{Ab} \quad \text{is right exact.}$$

Definition 3.2.5. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is called exact if it is both left exact and right exact.

Definition 3.2.6. Let M be an R -module.

1. M is called projective if the functor

$$\text{Hom}_R(M, -)$$

is exact.

2. M is called injective if the functor

$$\text{Hom}_R(-, M)$$

is exact.

3. M is called flat if the functor

$$M \otimes_R -$$

is exact.

If R is noncommutative, then in order for

$$M \otimes_R -$$

to be well-defined, M must be a right R -module.

3.3 Bimodules

Definition 3.3.1. Let R and S be rings. An abelian group M is called a left- R -right- S -bimodule if M is both a left R -module and a right S -module such that the two scalar multiplications satisfy

$$r(xs) = (rx)s \quad \text{for all } r \in R, s \in S, x \in M.$$

Notation.

${}_R M$ left R -module,

M_S right S -module,

${}_R M_S$ left R -right S -bimodule.

Let R, S be rings, and let ${}_R U_S$ be a bimodule.

For each left R -module ${}_R M$, there are two naturally induced S -module structures:

$\text{Hom}_R({}_R U_S, {}_R M)$, a left S -module,

$\text{Hom}_R({}_R M, {}_R U_S)$, a right S -module.

More precisely, for each $s \in S$:

if

$$f \in \text{Hom}_R({}_R U_S, {}_R M),$$

define

$$s \cdot f : U \rightarrow M, \quad u \mapsto f(us).$$

Similarly, if

$$f \in \text{Hom}_R({}_R M, {}_R U_S),$$

define

$$f \cdot s : M \rightarrow U, \quad m \mapsto f(m)s.$$

Thus we obtain functors

$$\mathrm{Hom}_R({}_R U_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod},$$

given on objects by

$${}_R M \longmapsto \mathrm{Hom}_R({}_R U_S, {}_R M),$$

which is covariant, and

$$\mathrm{Hom}_R(-, {}_R U_S) : R\text{-}\mathbf{Mod} \longrightarrow \mathbf{Mod}\text{-}S,$$

given on objects by

$${}_R M \longmapsto \mathrm{Hom}_R({}_R M, {}_R U_S),$$

which is contravariant.

On morphisms, for

$$f : M \rightarrow M',$$

the induced map

$$\mathrm{Hom}_R({}_R U_S, {}_R M) \longrightarrow \mathrm{Hom}_R({}_R U_S, {}_R M')$$

is given by

$$h \longmapsto f \circ h.$$

the induced map

$$\mathrm{Hom}_R({}_R M', {}_R U_S) \longrightarrow \mathrm{Hom}_R({}_R M, {}_R U_S)$$

is given by

$$h \longmapsto h \circ f.$$

Theorem 3.3.2. *The functor*

$$\mathrm{Hom}_R({}_R U_S, -) : R\text{-}\mathbf{Mod} \longrightarrow S\text{-}\mathbf{Mod}$$

is left exact.

Proof. Let

$$0 \longrightarrow N' \xrightarrow{f} N \xrightarrow{g} N''$$

be an exact sequence in $R\text{-}\mathbf{Mod}$. Consider the induced sequence

$$0 \longrightarrow \mathrm{Hom}_R({}_R U_S, N') \xrightarrow{\mathrm{Hom}_R(U, f)} \mathrm{Hom}_R({}_R U_S, N) \xrightarrow{\mathrm{Hom}_R(U, g)} \mathrm{Hom}_R({}_R U_S, N'').$$

We show that this sequence is exact.

First, $\mathrm{Hom}_R(U, f)$ is injective. Let

$$\varphi \in \mathrm{Hom}_R(U, N')$$

and suppose that

$$\text{Hom}_R(U, f)(\varphi) = f \circ \varphi = 0.$$

Then for every $u \in U$,

$$f(\varphi(u)) = 0.$$

Since f is injective, it follows that $\varphi(u) = 0$ for all $u \in U$. Hence $\varphi = 0$, so $\text{Hom}_R(U, f)$ is injective.

Next, we show that

$$\text{Im}(\text{Hom}_R(U, f)) \subseteq \ker(\text{Hom}_R(U, g)).$$

Indeed, if

$$\varphi \in \text{Hom}_R(U, N'),$$

then

$$\text{Hom}_R(U, g)(\text{Hom}_R(U, f)(\varphi)) = g \circ f \circ \varphi = 0,$$

because $g \circ f = 0$. Therefore

$$\text{Im}(\text{Hom}_R(U, f)) \subseteq \ker(\text{Hom}_R(U, g)).$$

Finally, let

$$\psi \in \ker(\text{Hom}_R(U, g)).$$

Then $\psi \in \text{Hom}_R(U, N)$ and

$$g \circ \psi = 0.$$

Thus for every $u \in U$,

$$g(\psi(u)) = 0,$$

so $\psi(u) \in \ker g = \text{Im } f$ by exactness. Hence for each $u \in U$ there exists some $y \in N'$ such that

$$f(y) = \psi(u).$$

Since f is injective, such y is unique. Define

$$\varphi : U \rightarrow N', \quad u \mapsto y,$$

where y is the unique element satisfying $f(y) = \psi(u)$.

Then

$$f \circ \varphi = \psi.$$

It remains to check that φ is R -linear. For $u, v \in U$,

$$f(\varphi(u + v)) = \psi(u + v) = \psi(u) + \psi(v) = f(\varphi(u)) + f(\varphi(v)) = f(\varphi(u) + \varphi(v)).$$

Since f is injective,

$$\varphi(u + v) = \varphi(u) + \varphi(v).$$

Similarly, for $r \in R$ and $u \in U$,

$$f(\varphi(ru)) = \psi(ru) = r\psi(u) = rf(\varphi(u)) = f(r\varphi(u)),$$

so again by injectivity of f ,

$$\varphi(ru) = r\varphi(u).$$

Thus $\varphi \in \text{Hom}_R(U, N')$, and

$$\text{Hom}_R(U, f)(\varphi) = f \circ \varphi = \psi.$$

Therefore

$$\ker(\text{Hom}_R(U, g)) \subseteq \text{Im}(\text{Hom}_R(U, f)).$$

Combining the two inclusions, we get

$$\text{Im}(\text{Hom}_R(U, f)) = \ker(\text{Hom}_R(U, g)).$$

Hence

$$0 \longrightarrow \text{Hom}_R({}_R U_S, N') \xrightarrow{\text{Hom}_R(U, f)} \text{Hom}_R({}_R U_S, N) \xrightarrow{\text{Hom}_R(U, g)} \text{Hom}_R({}_R U_S, N'')$$

is exact, and so the functor $\text{Hom}_R({}_R U_S, -)$ is left exact. $\square$

Theorem 3.3.3. *Let ${}_R U_S$ be an R - S -bimodule. Then the contravariant functor*

$$\text{Hom}_R(-, U) : R\text{-Mod} \longrightarrow \text{Mod-}S$$

is left exact.

Proof. Since $\text{Hom}_R(-, U)$ is contravariant, to say that it is left exact means the following:
for every short exact sequence of left R -modules

$$A \xrightarrow{f} B \xrightarrow{g} C \longrightarrow 0,$$

the induced sequence of right S -modules

$$0 \longrightarrow \text{Hom}_R(C, U) \xrightarrow{\text{Hom}_R(g, U)} \text{Hom}_R(B, U) \xrightarrow{\text{Hom}_R(f, U)} \text{Hom}_R(A, U)$$

is exact.

We verify this.

First, $\text{Hom}_R(g, U)$ is injective. Indeed, let $\varphi \in \text{Hom}_R(C, U)$ and suppose

$$\text{Hom}_R(g, U)(\varphi) = \varphi \circ g = 0.$$

Since g is surjective, it follows that $\varphi = 0$. Hence $\text{Hom}_R(g, U)$ is injective.

Next, we show

$$\text{Im}(\text{Hom}_R(g, U)) \subseteq \ker(\text{Hom}_R(f, U)).$$

Let $\varphi \in \text{Hom}_R(C, U)$. Then

$$\text{Hom}_R(f, U)(\text{Hom}_R(g, U)(\varphi)) = (\varphi \circ g) \circ f = \varphi \circ (g \circ f) = 0,$$

because $g \circ f = 0$. Therefore

$$\text{Im}(\text{Hom}_R(g, U)) \subseteq \ker(\text{Hom}_R(f, U)).$$

Finally, we prove the reverse inclusion. Let $\psi \in \text{Hom}_R(B, U)$ satisfy

$$\text{Hom}_R(f, U)(\psi) = \psi \circ f = 0.$$

Then ψ vanishes on $\text{Im}(f) = \ker(g)$. Hence ψ factors through the quotient

$$B/\ker(g).$$

Since g is surjective and $\ker(g) = \text{Im}(f)$, we have

$$B/\ker(g) \cong C.$$

Therefore there exists a unique R -module homomorphism $\bar{\psi} : C \rightarrow U$ such that

$$\psi = \bar{\psi} \circ g.$$

Thus

$$\psi \in \text{Im}(\text{Hom}_R(g, U)),$$

so

$$\ker(\text{Hom}_R(f, U)) \subseteq \text{Im}(\text{Hom}_R(g, U)).$$

Combining the two inclusions, we obtain

$$\text{Im}(\text{Hom}_R(g, U)) = \ker(\text{Hom}_R(f, U)).$$

Hence the sequence

$$0 \longrightarrow \text{Hom}_R(C, U) \xrightarrow{\text{Hom}_R(g, U)} \text{Hom}_R(B, U) \xrightarrow{\text{Hom}_R(f, U)} \text{Hom}_R(A, U)$$

is exact. This proves that $\text{Hom}_R(-, U)$ is left exact. $\square$

Let ${}_S M_R$ be a left- S -right- R -bimodule, ${}_R N$ be a left R -module, consider the tensor product $M \otimes_R N$.

For each $s \in S$, the mapping

$$M \times N \longrightarrow M \otimes_R N, \quad (m, n) \longmapsto (sm) \otimes n$$

is balanced.

So there is a unique abelian group homomorphism

$$u(s) : M \otimes_R N \longrightarrow M \otimes_R N$$

such that

$$u(s)(m \otimes n) = (sm) \otimes n.$$

Only need to check:

$$u : S \longrightarrow \text{End}_{\mathbb{Z}}(M \otimes_R N)$$

is a ring homomorphism.

If $M_R, {}_R N_T$, then $M \otimes_R N$ is a right T -module:

$$(m \otimes n)t = m \otimes (nt).$$

Lemma 3.3.4. *Let ${}_S M_R$ and ${}_R N_T$ be bimodules. Then $M \otimes_R N$ carries a natural structure of a left S -right T -bimodule, defined on elementary tensors by*

$$s(m \otimes n) = (sm) \otimes n, \quad (m \otimes n)t = m \otimes (nt),$$

for all $s \in S, t \in T, m \in M$, and $n \in N$.

Proof. Fix $s \in S$. Define

$$\beta_s : M \times N \rightarrow M \otimes_R N, \quad \beta_s(m, n) = sm \otimes n.$$

We claim that β_s is R -balanced. Indeed, it is additive in each variable, and for $r \in R$,

$$\beta_s(mr, n) = s(mr) \otimes n = (sm)r \otimes n = sm \otimes rn = \beta_s(m, rn).$$

Hence, by the universal property of the tensor product, there exists a unique group homomorphism

$$\lambda_s : M \otimes_R N \rightarrow M \otimes_R N$$

such that

$$\lambda_s(m \otimes n) = sm \otimes n.$$

We write $\lambda_s(x) = sx$.

Similarly, for fixed $t \in T$, define

$$\gamma_t : M \times N \rightarrow M \otimes_R N, \quad \gamma_t(m, n) = m \otimes nt.$$

Again γ_t is R -balanced, since for $r \in R$,

$$\gamma_t(mr, n) = mr \otimes nt = m \otimes r(nt) = m \otimes (rn)t = \gamma_t(m, rn).$$

Thus there exists a unique group homomorphism

$$\rho_t : M \otimes_R N \rightarrow M \otimes_R N$$

such that

$$\rho_t(m \otimes n) = m \otimes nt.$$

We write $\rho_t(x) = xt$.

It remains to verify the bimodule axioms. Since elementary tensors generate $M \otimes_R N$, it suffices to check them on $m \otimes n$:

$$(s_1 + s_2)(m \otimes n) = ((s_1 + s_2)m) \otimes n = (s_1m) \otimes n + (s_2m) \otimes n = s_1(m \otimes n) + s_2(m \otimes n),$$

$$(s_1s_2)(m \otimes n) = ((s_1s_2)m) \otimes n = s_1((s_2m) \otimes n) = s_1(s_2(m \otimes n)),$$

$$1_S(m \otimes n) = (1_Sm) \otimes n = m \otimes n,$$

and similarly

$$(m \otimes n)(t_1 + t_2) = m \otimes n(t_1 + t_2) = (m \otimes nt_1) + (m \otimes nt_2),$$

$$((m \otimes n)t_1)t_2 = (m \otimes nt_1)t_2 = m \otimes (nt_1)t_2 = m \otimes n(t_1t_2) = (m \otimes n)(t_1t_2),$$

$$(m \otimes n)1_T = m \otimes n.$$

Finally,

$$(s(m \otimes n))t = ((sm) \otimes n)t = (sm) \otimes nt = s(m \otimes nt) = s((m \otimes n)t).$$

Therefore $M \otimes_R N$ is a left S -right T -bimodule. □

Lemma 3.3.5. *Let M_R, M'_R be right R -modules, and let ${}_R N, {}_R N'$ be left R -modules. Suppose*

$$f_1, f_2, f \in \text{Hom}_R(M, M'), \quad g_1, g_2, g \in \text{Hom}_R(N, N').$$

Then for each pair (f, g) there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n) \quad (m \in M, n \in N).$$

Moreover,

1. $(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g);$
2. $f \otimes (g_1 + g_2) = (f \otimes g_1) + (f \otimes g_2);$
3. $f \otimes 0 = 0$ and $0 \otimes g = 0;$

$$4. 1_M \otimes 1_N = 1_{M \otimes_R N}.$$

Proof. Define

$$\beta : M \times N \rightarrow M' \otimes_R N', \quad \beta(m, n) = f(m) \otimes g(n).$$

We check that β is R -balanced. It is additive in each variable because f and g are module homomorphisms. Also, for $r \in R$,

$$\beta(mr, n) = f(mr) \otimes g(n) = f(m)r \otimes g(n) = f(m) \otimes rg(n) = f(m) \otimes g(rn) = \beta(m, rn).$$

Hence, by the universal property of $M \otimes_R N$, there exists a unique group homomorphism

$$f \otimes g : M \otimes_R N \rightarrow M' \otimes_R N'$$

such that

$$(f \otimes g)(m \otimes n) = f(m) \otimes g(n).$$

Now each asserted identity is verified on elementary tensors, and therefore on all of $M \otimes_R N$.

For (1), for all $m \in M$ and $n \in N$,

$$(((f_1 + f_2) \otimes g)(m \otimes n)) = (f_1 + f_2)(m) \otimes g(n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n),$$

while

$$((f_1 \otimes g) + (f_2 \otimes g))(m \otimes n) = (f_1 \otimes g)(m \otimes n) + (f_2 \otimes g)(m \otimes n) = f_1(m) \otimes g(n) + f_2(m) \otimes g(n).$$

Hence

$$(f_1 + f_2) \otimes g = (f_1 \otimes g) + (f_2 \otimes g).$$

The proof of (2) is analogous:

$$(f \otimes (g_1 + g_2))(m \otimes n) = f(m) \otimes (g_1 + g_2)(n) = f(m) \otimes g_1(n) + f(m) \otimes g_2(n),$$

which equals

$$((f \otimes g_1) + (f \otimes g_2))(m \otimes n).$$

For (3),

$$(f \otimes 0)(m \otimes n) = f(m) \otimes 0 = 0, \quad (0 \otimes g)(m \otimes n) = 0 \otimes g(n) = 0,$$

so both maps are the zero homomorphism.

For (4),

$$(1_M \otimes 1_N)(m \otimes n) = 1_M(m) \otimes 1_N(n) = m \otimes n,$$

hence $1_M \otimes 1_N$ is the identity on $M \otimes_R N$. □

Lemma 3.3.6. *Let M_R, M'_R, M''_R be right R -modules, and let ${}_R N, {}_R N', {}_R N''$ be left R -modules. Let*

$$f : M \rightarrow M', \quad f' : M' \rightarrow M'', \quad g : N \rightarrow N', \quad g' : N' \rightarrow N''$$

be R -module homomorphisms. Then

$$(f' \otimes g') \circ (f \otimes g) = (f' \circ f) \otimes (g' \circ g).$$

Proof. Again it suffices to check the equality on elementary tensors. For $m \in M$ and $n \in N$,

$$((f' \otimes g') \circ (f \otimes g))(m \otimes n) = (f' \otimes g')(f(m) \otimes g(n)) = f'(f(m)) \otimes g'(g(n)),$$

while

$$((f' \circ f) \otimes (g' \circ g))(m \otimes n) = (f' \circ f)(m) \otimes (g' \circ g)(n) = f'(f(m)) \otimes g'(g(n)).$$

Thus both homomorphisms agree on the generators $m \otimes n$ of $M \otimes_R N$, hence they are equal. $\square$

Definition 3.3.7. *Let ${}_S U_R$ be a bimodule. Define*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

by

$$M \longmapsto U \otimes_R M, \quad f \longmapsto \text{id}_U \otimes f.$$

Similarly, Let ${}_R U_S$ be a bimodule.

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

by

$$N \longmapsto N \otimes_R U, \quad g \longmapsto g \otimes \text{id}_U.$$

Proposition 3.3.8. *Let ${}_S U_R$ be a bimodule. Then*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is a covariant functor.

Similarly, if ${}_R U_S$ is a bimodule, then

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

is a covariant functor.

Proof. We prove the first statement. For each left R -module M , the tensor product

$$U \otimes_R M$$

is naturally a left S -module via

$$s(u \otimes m) = (su) \otimes m \quad (s \in S, u \in U, m \in M).$$

For each R -module homomorphism $f : M \rightarrow N$, define

$$U \otimes_R f : U \otimes_R M \rightarrow U \otimes_R N$$

by

$$(U \otimes_R f)(u \otimes m) = u \otimes f(m).$$

Equivalently,

$$U \otimes_R f = \text{id}_U \otimes f.$$

This is well-defined and is an S -module homomorphism.

It remains to verify the functorial properties. For the identity map $\text{id}_M : M \rightarrow M$, we have

$$U \otimes_R \text{id}_M = \text{id}_U \otimes \text{id}_M = \text{id}_{U \otimes_R M}.$$

If

$$M \xrightarrow{f} N \xrightarrow{g} P$$

are R -module homomorphisms, then for every $u \otimes m \in U \otimes_R M$,

$$(U \otimes_R (g \circ f))(u \otimes m) = u \otimes (g \circ f)(m) = g(u \otimes f(m)) = ((U \otimes_R g) \circ (U \otimes_R f))(u \otimes m).$$

Hence

$$U \otimes_R (g \circ f) = (U \otimes_R g) \circ (U \otimes_R f).$$

Therefore $U \otimes_R -$ is a covariant functor from $R\text{-Mod}$ to $S\text{-Mod}$.

The proof of the second statement is entirely similar. If ${}_R U_S$ is a bimodule and N is a right R -module, then

$$N \otimes_R U$$

is naturally a right S -module via

$$(n \otimes u)s = n \otimes (us).$$

For a homomorphism $g : N \rightarrow N'$ in $\text{Mod-}R$, define

$$g \otimes_R U = g \otimes \text{id}_U : N \otimes_R U \rightarrow N' \otimes_R U$$

by

$$(g \otimes_R U)(n \otimes u) = g(n) \otimes u.$$

One checks immediately that identities and compositions are preserved. Thus

$$- \otimes_R U : \text{Mod-}R \rightarrow \text{Mod-}S$$

is also a covariant functor. □

Theorem 3.3.9. *Let ${}_S U_R$ be a bimodule. Then the functor*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is right exact. In other words, if

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is an exact sequence of left R -modules, then

$$U \otimes_R N' \xrightarrow{\text{id}_U \otimes f} U \otimes_R N \xrightarrow{\text{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact.

Proof. Since

$$N' \xrightarrow{f} N \xrightarrow{g} N'' \longrightarrow 0$$

is exact, we have

$$\ker g = \text{Im } f \quad \text{and} \quad N'' \cong N / \text{Im } f.$$

Let

$$\pi : N \longrightarrow N / \text{Im } f$$

be the canonical projection. Then there exists an isomorphism

$$\alpha : N / \text{Im } f \longrightarrow N''$$

such that

$$g = \alpha \circ \pi.$$

Now consider the map

$$\Phi : U \otimes_R N \longrightarrow U \otimes_R (N / \text{Im } f), \quad u \otimes n \longmapsto u \otimes \pi(n).$$

By the universal property of tensor products, Φ is a well-defined S -module homomorphism.

We claim that

$$\ker \Phi = \text{Im}(\text{id}_U \otimes f).$$

First, since $\pi \circ f = 0$, functoriality gives

$$\Phi \circ (\text{id}_U \otimes f) = \text{id}_U \otimes (\pi \circ f) = 0.$$

Hence

$$\text{Im}(\text{id}_U \otimes f) \subseteq \ker \Phi.$$

To prove the reverse inclusion, define

$$\overline{\Phi} : (U \otimes_R N) / \text{Im}(\text{id}_U \otimes f) \longrightarrow U \otimes_R (N / \text{Im } f)$$

by

$$\overline{\Phi}((u \otimes n) + \text{Im}(\text{id}_U \otimes f)) = u \otimes \pi(n).$$

This is well defined because Φ vanishes on $\text{Im}(\text{id}_U \otimes f)$.

Next define

$$\Psi : U \times (N / \text{Im } f) \longrightarrow (U \otimes_R N) / \text{Im}(\text{id}_U \otimes f)$$

by

$$\Psi(u, \bar{n}) = (u \otimes n) + \text{Im}(\text{id}_U \otimes f),$$

where $\bar{n} = n + \text{Im } f$.

We check that Ψ is well defined. If $\bar{n} = \bar{n}'$, then

$$n - n' \in \text{Im } f,$$

so there exists $x \in N'$ such that

$$n - n' = f(x).$$

Hence

$$(u \otimes n) - (u \otimes n') = u \otimes (n - n') = u \otimes f(x) = (\text{id}_U \otimes f)(u \otimes x) \in \text{Im}(\text{id}_U \otimes f).$$

Thus

$$(u \otimes n) + \text{Im}(\text{id}_U \otimes f) = (u \otimes n') + \text{Im}(\text{id}_U \otimes f).$$

So Ψ is well defined. It is clearly R -balanced, hence induces an S -module homomorphism

$$\overline{\Psi} : U \otimes_R (N / \text{Im } f) \longrightarrow (U \otimes_R N) / \text{Im}(\text{id}_U \otimes f)$$

such that

$$\overline{\Psi}(u \otimes \bar{n}) = (u \otimes n) + \text{Im}(\text{id}_U \otimes f).$$

It is immediate that $\overline{\Phi}$ and $\overline{\Psi}$ are inverse to each other. Therefore

$$(U \otimes_R N) / \text{Im}(\text{id}_U \otimes f) \cong U \otimes_R (N / \text{Im } f).$$

It follows that

$$\ker \Phi = \text{Im}(\text{id}_U \otimes f).$$

Finally, since $\alpha : N / \text{Im } f \rightarrow N''$ is an isomorphism, the induced map

$$\text{id}_U \otimes \alpha : U \otimes_R (N / \text{Im } f) \longrightarrow U \otimes_R N''$$

is an isomorphism. Moreover,

$$\mathrm{id}_U \otimes g = (\mathrm{id}_U \otimes \alpha) \circ \Phi.$$

Hence

$$\ker(\mathrm{id}_U \otimes g) = \ker \Phi = \mathrm{Im}(\mathrm{id}_U \otimes f),$$

and $\mathrm{id}_U \otimes g$ is surjective because Φ is surjective and $\mathrm{id}_U \otimes \alpha$ is an isomorphism.

Therefore

$$U \otimes_R N' \xrightarrow{\mathrm{id}_U \otimes f} U \otimes_R N \xrightarrow{\mathrm{id}_U \otimes g} U \otimes_R N'' \longrightarrow 0$$

is exact. □

Chapter 4

Lecture 4: Projectives, Injectives

Theorem 4.0.1. *Let ${}_S U_R$ be a left- S -right- R -bimodule. Then the functor*

$$U \otimes_R - : R\text{-Mod} \longrightarrow S\text{-Mod}$$

is right exact.

Proof. Let

$$M' \xrightarrow{f} M \xrightarrow{g} M'' \rightarrow 0$$

be an exact sequence of left R -modules. We must show that

$$U \otimes_R M' \xrightarrow{1_U \otimes f} U \otimes_R M \xrightarrow{1_U \otimes g} U \otimes_R M'' \rightarrow 0$$

is exact in $S\text{-Mod}$.

First, $1_U \otimes g$ is surjective. Indeed, every element of $U \otimes_R M''$ is a finite sum of elementary tensors of the form $u \otimes m''$, with $u \in U$ and $m'' \in M''$. Since g is surjective, for each $m'' \in M''$ there exists $m \in M$ such that $g(m) = m''$. Hence

$$u \otimes m'' = u \otimes g(m) = (1_U \otimes g)(u \otimes m),$$

so $1_U \otimes g$ is surjective.

Next, since $g \circ f = 0$, we have

$$(1_U \otimes g) \circ (1_U \otimes f) = 1_U \otimes (g \circ f) = 0.$$

Therefore,

$$\text{im}(1_U \otimes f) \subseteq \ker(1_U \otimes g).$$

It remains to prove the reverse inclusion. Consider the quotient module

$$(U \otimes_R M) / \text{im}(1_U \otimes f).$$

Define a map

$$\Phi : U \times M'' \longrightarrow (U \otimes_R M) / \text{im}(1_U \otimes f)$$

by

$$\Phi(u, m'') = \overline{u \otimes m},$$

where $m \in M$ is any element such that $g(m) = m''$.

We first check that Φ is well defined. Suppose $m_1, m_2 \in M$ both satisfy $g(m_1) = g(m_2) = m''$. Then

$$m_1 - m_2 \in \ker g = \operatorname{im} f,$$

so there exists $m' \in M'$ such that $m_1 - m_2 = f(m')$. Hence

$$u \otimes m_1 - u \otimes m_2 = u \otimes (m_1 - m_2) = u \otimes f(m') = (1_U \otimes f)(u \otimes m'),$$

which lies in $\operatorname{im}(1_U \otimes f)$. Thus

$$\overline{u \otimes m_1} = \overline{u \otimes m_2},$$

so Φ is well defined.

It is straightforward to verify that Φ is R -balanced, hence by the universal property of the tensor product there exists a unique S -module homomorphism

$$\overline{\Phi} : U \otimes_R M'' \longrightarrow (U \otimes_R M) / \operatorname{im}(1_U \otimes f)$$

such that

$$\overline{\Phi}(u \otimes m'') = \overline{u \otimes m}, \quad g(m) = m''.$$

Now define

$$\Psi : U \otimes_R M \longrightarrow U \otimes_R M'', \quad \Psi(u \otimes m) = u \otimes g(m).$$

This is exactly $1_U \otimes g$. Since

$$\operatorname{im}(1_U \otimes f) \subseteq \ker(1_U \otimes g),$$

the map Ψ factors uniquely through the quotient, giving an S -module homomorphism

$$\tilde{\Psi} : (U \otimes_R M) / \operatorname{im}(1_U \otimes f) \longrightarrow U \otimes_R M''.$$

We now show that $\overline{\Phi}$ and $\tilde{\Psi}$ are inverse to each other. For $u \otimes m'' \in U \otimes_R M''$, choose $m \in M$ with $g(m) = m''$. Then

$$(\tilde{\Psi} \circ \overline{\Phi})(u \otimes m'') = \tilde{\Psi}(\overline{u \otimes m}) = u \otimes g(m) = u \otimes m''.$$

Conversely, for $\overline{u \otimes m} \in (U \otimes_R M) / \operatorname{im}(1_U \otimes f)$,

$$(\overline{\Phi} \circ \tilde{\Psi})(\overline{u \otimes m}) = \overline{\Phi}(u \otimes g(m)) = \overline{u \otimes m}.$$

Thus $\overline{\Phi}$ is an isomorphism.

Therefore,

$$(U \otimes_R M) / \operatorname{im}(1_U \otimes f) \cong U \otimes_R M'',$$

and hence

$$\ker(1_U \otimes g) = \text{im}(1_U \otimes f).$$

So the sequence

$$U \otimes_R M' \xrightarrow{1_U \otimes f} U \otimes_R M \xrightarrow{1_U \otimes g} U \otimes_R M'' \rightarrow 0$$

is exact. This proves that the functor $U \otimes_R -$ is right exact. $\square$

Theorem 4.0.2. *Let ${}_R U_S$ be a left- R -right- S -bimodule. Then the functor*

$$- \otimes_R U : \text{Mod-}R \longrightarrow \text{Mod-}S$$

is right exact.

4.1 Projective, Injective and Flat Modules

Definition 4.1.1. *Let R be a ring.*

- *A left R -module P is called projective if the functor*

$$\text{Hom}_R(P, -) : R\text{-Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R -module I is called injective if the contravariant functor*

$$\text{Hom}_R(-, I) : R\text{-Mod} \longrightarrow \mathbf{Ab}$$

is exact.

- *A left R -module F is called flat if the functor*

$$- \otimes_R F : \text{Mod-}R \longrightarrow \mathbf{Ab}$$

is exact.

Theorem 4.1.2. *Let ${}_R P$ be a left R -module. The following statements are equivalent.*

1. *Whenever we are given a commutative diagram*

$$\begin{array}{ccc} & P & \\ \tilde{\gamma} \swarrow & \downarrow \gamma & \\ M & \xrightarrow[g]{} N & \longrightarrow 0 \end{array}$$

with g an epimorphism, there exists an R -homomorphism

$$\tilde{\gamma} : P \rightarrow M$$

such that $g \circ \tilde{\gamma} = \gamma$.

2. For every epimorphism $f : M \rightarrow N$ of left R -modules, the induced map

$$\text{Hom}_R(P, f) : \text{Hom}_R(P, M) \longrightarrow \text{Hom}_R(P, N)$$

is surjective.

3. For every ring S and every right S -module structure on P making ${}_R P_S$ an R - S -bimodule, the functor

$$\text{Hom}_R(P, -) : {}_R \text{Mod} \longrightarrow \text{Mod-}S$$

is exact.

4. For every short exact sequence

$$0 \longrightarrow M' \xrightarrow{u} M \xrightarrow{v} M'' \longrightarrow 0$$

of left R -modules, the sequence

$$0 \longrightarrow \text{Hom}_R(P, M') \xrightarrow{\text{Hom}_R(P, u)} \text{Hom}_R(P, M) \xrightarrow{\text{Hom}_R(P, v)} \text{Hom}_R(P, M'') \longrightarrow 0$$

is exact.

Proof. (1) $\iff$ (2). Let $f : M \rightarrow N$ be an epimorphism. A morphism $\gamma : P \rightarrow N$ admits a lift $\tilde{\gamma} : P \rightarrow M$ with $f\tilde{\gamma} = \gamma$ if and only if γ lies in the image of

$$\text{Hom}_R(P, f) : \text{Hom}_R(P, M) \rightarrow \text{Hom}_R(P, N).$$

Thus (1) and (2) are equivalent.

(3) $\implies$ (4). This is immediate by taking the codomain category to be **Ab**.

(4) $\implies$ (2). Apply (4) to the short exact sequence

$$0 \longrightarrow \ker f \longrightarrow M \xrightarrow{f} N \longrightarrow 0.$$

Then $\text{Hom}_R(P, f)$ is surjective.

(2) $\implies$ (3). By the left exactness of $\text{Hom}_R(P, -)$, it remains only to prove surjectivity at the last term in every short exact sequence. But this is exactly the content of (2). $\square$

Proposition 4.1.3. *Free modules are projective.*

Proof. Let $F = \bigoplus_{i \in I} Re_i$ be a free left R -module with basis $\{e_i\}_{i \in I}$. Let $g : M \rightarrow N$ be an epimorphism and let $\gamma : F \rightarrow N$ be an R -homomorphism. For each $i \in I$, choose $m_i \in M$ such that

$$g(m_i) = \gamma(e_i).$$

Since F is free, there exists a unique R -homomorphism

$$\tilde{\gamma} : F \rightarrow M$$

such that $\tilde{\gamma}(e_i) = m_i$ for all $i \in I$. Then

$$g\tilde{\gamma}(e_i) = g(m_i) = \gamma(e_i)$$

for every basis element, so $g\tilde{\gamma} = \gamma$. Hence F is projective. $\square$

Theorem 4.1.4. *Let ${}_R P$ be a left R -module. The following statements are equivalent.*

1. P is projective.

2. Every epimorphism

$$f : M \longrightarrow P \longrightarrow 0$$

splits, that is, there exists an R -homomorphism $g : P \rightarrow M$ such that

$$f \circ g = \text{id}_P.$$

3. P is isomorphic to a direct summand of a free module.

Proof. (1) $\implies$ (2). Apply the lifting property to the diagram

$$\begin{array}{ccc} & P & \\ g \swarrow & \downarrow \text{id}_P & \\ M & \xrightarrow{f} P & \longrightarrow 0 \end{array}$$

to obtain $f \circ g = \text{id}_P$.

(2) $\implies$ (3). Every module is a homomorphic image of a free module, so choose an epimorphism

$$\pi : F \rightarrow P$$

with F free. By (2), π splits, so there exists $s : P \rightarrow F$ such that

$$\pi \circ s = \text{id}_P.$$

Therefore

$$F \cong \ker \pi \oplus s(P) \cong \ker \pi \oplus P,$$

and hence P is a direct summand of the free module F .

(3) $\implies$ (1). Write

$$F \cong P \oplus Q$$

with F free. Since free modules are projective, it suffices to show that a direct summand of a projective module is projective. Let $f : M \rightarrow N$ be an epimorphism and

$$\alpha : P \rightarrow N$$

an R -homomorphism. Define

$$\hat{\alpha} : P \oplus Q \rightarrow N, \quad \hat{\alpha}(p, q) = \alpha(p).$$

Because F is projective, there exists

$$\hat{\beta} : P \oplus Q \rightarrow M$$

such that $f \circ \hat{\beta} = \hat{\alpha}$. Let

$$\iota : P \rightarrow P \oplus Q, \quad p \mapsto (p, 0).$$

Then

$$f \circ (\hat{\beta} \circ \iota) = \hat{\alpha} \circ \iota = \alpha.$$

Hence P is projective. □

Corollary 4.1.5. *A left R -module P is finitely generated and projective if and only if for some R -module P' and some integer $n \geq 0$ there is an R -isomorphism*

$$P \oplus P' \cong R^n.$$

Proof. If $P \oplus P' \cong R^n$, then P is a direct summand of a finite free module, hence finitely generated and projective.

Conversely, assume that P is finitely generated and projective. By the theorem above, there exists a free module

$$F = \bigoplus_{i \in I} Re_i$$

and an R -module Q such that

$$F \cong P \oplus Q.$$

Let $p : F \rightarrow P$ and $s : P \rightarrow F$ be the projection and inclusion corresponding to this direct sum decomposition. Choose generators $x_1, \dots, x_m$ of P . Each $s(x_j)$ is a finite linear combination of the basis elements $\{e_i\}_{i \in I}$, so there exists a finite subset $J \subseteq I$ such that

$$s(x_1), \dots, s(x_m) \in F_J := \bigoplus_{j \in J} Re_j.$$

Since the x_j generate P , we have $s(P) \subseteq F_J$. Hence the restriction

$$p|_{F_J} : F_J \rightarrow P$$

is surjective, and the map $s : P \rightarrow F_J$ is a section of $p|_{F_J}$. Therefore

$$F_J \cong P \oplus \ker(p|_{F_J}).$$

Since $F_J \cong R^{|J|}$, the result follows. □

4.2 Schanuel Lemma

Theorem 4.2.1 (Schanuel Lemma). *Let*

$$0 \longrightarrow K \longrightarrow P \xrightarrow{\lambda} M \longrightarrow 0$$

and

$$0 \longrightarrow K' \longrightarrow P' \xrightarrow{\lambda'} M \longrightarrow 0$$

be short exact sequences of left R -modules, with P and P' projective. Then

$$K' \oplus P \cong K \oplus P'.$$

Proof. Consider the pullback

$$Y := \{(p, p') \in P \oplus P' \mid \lambda(p) = \lambda'(p')\}.$$

There are natural exact sequences

$$0 \longrightarrow K \longrightarrow Y \xrightarrow{\mu'} P' \longrightarrow 0, \quad \mu'(p, p') = p',$$

and

$$0 \longrightarrow K' \longrightarrow Y \xrightarrow{\mu} P \longrightarrow 0, \quad \mu(p, p') = p.$$

Since P and P' are projective, both sequences split. Therefore

$$Y \cong K \oplus P' \quad \text{and} \quad Y \cong K' \oplus P.$$

Hence

$$K \oplus P' \cong K' \oplus P.$$

□

Lemma 4.2.2. *Let M be a left R -module. Then*

$$\text{Hom}_R({}_R R, M) \cong M$$

as left R -modules.

Proof. Define

$$\Phi : \text{Hom}_R({}_R R, M) \rightarrow M, \quad f \mapsto f(1).$$

This map is R -linear. Define also

$$\Psi : M \rightarrow \text{Hom}_R({}_R R, M), \quad m \mapsto (r \mapsto rm).$$

For $f \in \text{Hom}_R({}_R R, M)$ and $r \in R$,

$$\Psi(\Phi(f))(r) = rf(1) = f(r \cdot 1) = f(r),$$

so $\Psi\Phi = \text{id}$. Also

$$\Phi(\Psi(m)) = \Psi(m)(1) = 1m = m,$$

hence Φ and Ψ are inverse isomorphisms. $\square$

Corollary 4.2.3. *For any ring R with 1,*

$$\text{End}_R({}_R R) \cong R^{\text{op}}$$

as rings.

Proof. By the lemma, every $f \in \text{End}_R({}_R R)$ is determined by $f(1)$, and

$$f(r) = rf(1)$$

for all $r \in R$. Thus

$$\Theta : \text{End}_R({}_R R) \rightarrow R^{\text{op}}, \quad f \mapsto f(1)$$

is bijective. Moreover, for $f, g \in \text{End}_R({}_R R)$,

$$(f \circ g)(1) = f(g(1)) = g(1)f(1),$$

which is multiplication in the opposite ring. Therefore Θ is a ring isomorphism. $\square$

4.3 Projective Morphisms

Lemma 4.3.1. *Let R be a commutative ring with 1, and let X, Y, M be R -modules. Then there is an R -module homomorphism*

$$\Theta_M^{X,Y} : \text{Hom}_R(X, M) \otimes_R \text{Hom}_R(M, Y) \longrightarrow \text{Hom}_R(X, Y), \quad \varphi \otimes \psi \longmapsto \psi \circ \varphi.$$

Its image consists exactly of those morphisms from X to Y which factor through some finite direct power M^n of M .

Proof. The map is well defined and R -bilinear because R is commutative, so it induces an R -linear map on the tensor product.

Let

$$f = \Theta_M^{X,Y} \left(\sum_{i=1}^n \varphi_i \otimes \psi_i \right) = \sum_{i=1}^n \psi_i \circ \varphi_i.$$

Define

$$\alpha : X \rightarrow M^n, \quad \alpha(x) = (\varphi_1(x), \dots, \varphi_n(x)),$$

and

$$\beta : M^n \rightarrow Y, \quad \beta(m_1, \dots, m_n) = \sum_{i=1}^n \psi_i(m_i).$$

Then

$$f = \beta \circ \alpha,$$

so every element of the image factors through M^n for some n .

Conversely, suppose

$$f : X \xrightarrow{u} M^n \xrightarrow{v} Y$$

factors through M^n . Let $\pi_i : M^n \rightarrow M$ be the i th projection and $\iota_i : M \rightarrow M^n$ the i th inclusion. Then

$$\text{id}_{M^n} = \sum_{i=1}^n \iota_i \pi_i,$$

hence

$$f = v \circ u = v \circ \left(\sum_{i=1}^n \iota_i \pi_i \right) \circ u = \sum_{i=1}^n (v \circ \iota_i) \circ (\pi_i \circ u).$$

Therefore

$$f = \Theta_M^{X,Y} \left(\sum_{i=1}^n (\pi_i \circ u) \otimes (v \circ \iota_i) \right),$$

and so f belongs to the image. □

Definition 4.3.2. Let R be a commutative ring with 1. For an R -module X , write

$$X^* := \text{Hom}_R(X, R).$$

For R -modules X and Y , define

$$\tau_{X,Y} : X^* \otimes_R Y \longrightarrow \text{Hom}_R(X, Y), \quad \varphi \otimes y \longmapsto (x \mapsto \varphi(x)y).$$

A morphism $f : X \rightarrow Y$ is called a projective morphism if it factors through a finitely generated projective module.

Proposition 4.3.3. Let R be a commutative ring with 1. A morphism $f : X \rightarrow Y$ is projective if and only if

$$f \in \text{Im}(\tau_{X,Y}).$$

Proof. By the previous lemma with $M = R$, the image of $\tau_{X,Y}$ consists exactly of those morphisms that factor through some finite free module R^n .

Since finite free modules are finitely generated projective, every morphism in $\text{Im}(\tau_{X,Y})$ is projective.

Conversely, suppose $f : X \rightarrow Y$ factors through a finitely generated projective module P :

$$X \xrightarrow{u} P \xrightarrow{v} Y.$$

Choose Q and n such that

$$P \oplus Q \cong R^n.$$

Let $i : P \rightarrow R^n$ and $p : R^n \rightarrow P$ be the inclusion and projection corresponding to this decomposition, so that $p \circ i = \text{id}_P$. Then

$$f = v \circ p \circ i \circ u,$$

hence f factors through R^n . Therefore $f \in \text{Im}(\tau_{X,Y})$. $\square$

Definition 4.3.4. For R -modules X and Y , we write

$$\text{Hom}_R^{\text{pr}}(X, Y)$$

for the set of projective morphisms from X to Y .

Lemma 4.3.5. For any R -module M , the set

$$\text{Hom}_R^{\text{pr}}(M, M)$$

is a two-sided ideal of the ring $\text{End}_R(M)$.

Proof. Let $f \in \text{Hom}_R^{\text{pr}}(M, M)$. Then f factors through a finitely generated projective module P :

$$M \xrightarrow{u} P \xrightarrow{v} M.$$

For any $a, b \in \text{End}_R(M)$,

$$a \circ f \circ b = (a \circ v) \circ u \circ b$$

still factors through P . Hence $a \circ f$ and $f \circ b$ are projective morphisms, so $\text{Hom}_R^{\text{pr}}(M, M)$ is a two-sided ideal of $\text{End}_R(M)$. $\square$

Theorem 4.3.6. Let R be a commutative ring with 1, and let P be an R -module. The following statements are equivalent.

1. P is a finitely generated projective module.
- 2.

$$\text{Hom}_R^{\text{pr}}(P, P) = \text{End}_R(P).$$

3. For every R -module X , the morphism

$$\tau_{X,P} : X^* \otimes_R P \longrightarrow \text{Hom}_R(X, P)$$

is an isomorphism.

4. For every R -module X , the morphism

$$\tau_{P,X} : P^* \otimes_R X \longrightarrow \text{Hom}_R(P, X)$$

is an isomorphism.

5. The morphism

$$\tau_{P,P} : P^* \otimes_R P \longrightarrow \text{End}_R(P)$$

is an isomorphism.

Proof. (1) $\implies$ (2). Since P itself is finitely generated projective, the identity morphism

$$\text{id}_P : P \rightarrow P$$

is projective. By the previous lemma, $\text{Hom}_R^{\text{pr}}(P, P)$ is a two-sided ideal of $\text{End}_R(P)$ containing id_P , and therefore equals the whole endomorphism ring.

(3) $\implies$ (5) and (4) $\implies$ (5) are immediate by taking $X = P$.

(2) $\implies$ (3). By (2), id_P is projective, so there exist $\varphi_1, \dots, \varphi_n \in P^*$ and $p_1, \dots, p_n \in P$ such that

$$\text{id}_P = \tau_{P,P} \left(\sum_{i=1}^n \varphi_i \otimes p_i \right).$$

Equivalently,

$$p = \sum_{i=1}^n \varphi_i(p) p_i \quad \text{for all } p \in P.$$

For any R -module X , define

$$\sigma_X : \text{Hom}_R(X, P) \rightarrow X^* \otimes_R P, \quad \alpha \mapsto \sum_{i=1}^n (\varphi_i \circ \alpha) \otimes p_i.$$

Then for $\alpha \in \text{Hom}_R(X, P)$ and $x \in X$,

$$\tau_{X,P}(\sigma_X(\alpha))(x) = \sum_{i=1}^n (\varphi_i \circ \alpha)(x) p_i = \sum_{i=1}^n \varphi_i(\alpha(x)) p_i = \alpha(x),$$

so $\tau_{X,P} \circ \sigma_X = \text{id}$.

Conversely, let

$$\xi = \sum_{j=1}^m \psi_j \otimes q_j \in X^* \otimes_R P.$$

Then

$$\begin{aligned}
\sigma_X(\tau_{X,P}(\xi)) &= \sum_{i=1}^n \left(\varphi_i \circ \tau_{X,P}(\xi) \right) \otimes p_i \\
&= \sum_{i=1}^n \left(x \mapsto \varphi_i \left(\sum_{j=1}^m \psi_j(x) q_j \right) \right) \otimes p_i \\
&= \sum_{i=1}^n \sum_{j=1}^m (\psi_j \varphi_i(q_j)) \otimes p_i \\
&= \sum_{j=1}^m \psi_j \otimes \left(\sum_{i=1}^n \varphi_i(q_j) p_i \right) \\
&= \sum_{j=1}^m \psi_j \otimes q_j \\
&= \xi.
\end{aligned}$$

Therefore σ_X is the inverse of $\tau_{X,P}$, and $\tau_{X,P}$ is an isomorphism.

(2) $\implies$ (4). Using the same decomposition of id_P , define for any R -module X

$$\rho_X : \text{Hom}_R(P, X) \rightarrow P^* \otimes_R X, \quad \beta \mapsto \sum_{i=1}^n \varphi_i \otimes \beta(p_i).$$

For $\beta \in \text{Hom}_R(P, X)$ and $p \in P$,

$$\tau_{P,X}(\rho_X(\beta))(p) = \sum_{i=1}^n \varphi_i(p) \beta(p_i) = \beta \left(\sum_{i=1}^n \varphi_i(p) p_i \right) = \beta(p),$$

so $\tau_{P,X} \rho_X = \text{id}$. Conversely, for

$$\eta = \sum_{j=1}^m \varphi'_j \otimes x_j \in P^* \otimes_R X,$$

we have

$$\begin{aligned}
\rho_X(\tau_{P,X}(\eta)) &= \sum_{i=1}^n \varphi_i \otimes \tau_{P,X}(\eta)(p_i) \\
&= \sum_{i=1}^n \varphi_i \otimes \left(\sum_{j=1}^m \varphi'_j(p_i) x_j \right) \\
&= \sum_{j=1}^m \left(\sum_{i=1}^n \varphi'_j(p_i) \varphi_i \right) \otimes x_j \\
&= \sum_{j=1}^m \varphi'_j \otimes x_j \\
&= \eta,
\end{aligned}$$

because

$$\varphi'_j(p) = \varphi'_j\left(\sum_{i=1}^n \varphi_i(p)p_i\right) = \sum_{i=1}^n \varphi_i(p)\varphi'_j(p_i)$$

for all $p \in P$. Hence $\tau_{P,X}$ is an isomorphism.

(5) $\implies$ (1). Since $\tau_{P,P}$ is surjective, we can write

$$\text{id}_P = \tau_{P,P}\left(\sum_{i=1}^n \varphi_i \otimes p_i\right)$$

for some $\varphi_i \in P^*$ and $p_i \in P$. Define

$$\iota : P \rightarrow R^n, \quad \iota(p) = (\varphi_1(p), \dots, \varphi_n(p)),$$

and

$$\pi : R^n \rightarrow P, \quad \pi(a_1, \dots, a_n) = \sum_{i=1}^n a_i p_i.$$

Then

$$(\pi \circ \iota)(p) = \sum_{i=1}^n \varphi_i(p)p_i = p$$

for all $p \in P$. Thus $\pi \circ \iota = \text{id}_P$, so P is a direct summand of the finite free module R^n . Therefore P is finitely generated and projective. $\square$

4.4 Projective Covers

Definition 4.4.1. Let R be a ring with 1. An epimorphism

$$f : U \rightarrow V$$

is called an *essential epimorphism* if no proper submodule of U is mapped surjectively onto V by f . Equivalently, whenever

$$g : W \rightarrow U$$

is a morphism such that $f \circ g$ is an epimorphism, then g is an epimorphism.

Proof. Suppose first that no proper submodule of U maps onto V . Let $g : W \rightarrow U$ satisfy $f \circ g$ epimorphic. Then

$$f(g(W)) = V.$$

By assumption, $g(W)$ cannot be a proper submodule of U , so $g(W) = U$ and g is surjective.

Conversely, assume that whenever $f \circ g$ is epi, then g is epi. Let $U_0 \subsetneq U$ be a proper submodule and let

$$\iota : U_0 \hookrightarrow U$$

be the inclusion. If $f(U_0) = V$, then $f \circ \iota$ is epi, hence ι is epi, which is impossible. Thus

$f(U_0) \neq V$. □

Definition 4.4.2. A projective cover of an R -module U is an essential epimorphism

$$P \rightarrow U$$

with P projective.

Proposition 4.4.3. Let

$$U \xrightarrow{f} V \xrightarrow{g} W$$

be R -homomorphisms. If any two of f , g , and $g \circ f$ are essential epimorphisms, then so is the third.

Proof. If f and g are essential, then $g \circ f$ is essential. Let $U_0 \subsetneq U$ be a proper submodule. Since f is essential,

$$f(U_0) \neq V.$$

As g is essential, $g(f(U_0)) \neq W$. Hence $(g \circ f)(U_0) \neq W$, so $g \circ f$ is essential.

If g and $g \circ f$ are essential, then f is essential. Since $g \circ f$ is epic, f is epic. Let $U_0 \subsetneq U$ be proper. If $f(U_0) = V$, then

$$(g \circ f)(U_0) = W,$$

contradicting that $g \circ f$ is essential. Thus $f(U_0) \neq V$, so f is essential.

If f and $g \circ f$ are essential, then g is essential. Since $g \circ f$ is epic and f is epic, g is epic. Let $V_0 \subsetneq V$ be a proper submodule, and set

$$U_0 := f^{-1}(V_0).$$

Because f is surjective, U_0 is a proper submodule of U . Since $g \circ f$ is essential,

$$(g \circ f)(U_0) \neq W.$$

But

$$(g \circ f)(U_0) = g(V_0),$$

so $g(V_0) \neq W$. Hence g is essential. □

4.5 Noetherian and Artinian Modules

Definition 4.5.1. An R -module M is called Noetherian if for every ascending chain of submodules

$$L_1 \subseteq L_2 \subseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \quad \text{for all } n \geq n_0.$$

Lemma 4.5.2. *For an R -module M , the following are equivalent.*

1. M is Noetherian.
2. Every submodule of M is finitely generated.
3. Every non-empty set of submodules of M has a maximal element.

Proof. **(1) $\implies$ (3).** Suppose $\mathcal{S}$ is a non-empty set of submodules of M having no maximal element. Choose $L_1 \in \mathcal{S}$. Since L_1 is not maximal, there exists $L_2 \in \mathcal{S}$ with

$$L_1 \subsetneq L_2.$$

Continuing inductively, we obtain a strictly ascending chain

$$L_1 \subsetneq L_2 \subsetneq L_3 \subsetneq \cdots,$$

contradicting that M is Noetherian.

(3) $\implies$ (2). Let $N \leq M$. Consider the set

$$\mathcal{T} := \{L \leq N \mid L \text{ is finitely generated}\}.$$

This set is non-empty because $0 \in \mathcal{T}$. By (3), $\mathcal{T}$ has a maximal element, say L . If $L \neq N$, choose $x \in N \setminus L$. Then

$$L + Rx$$

is a finitely generated submodule of N strictly containing L , contradicting the maximality of L . Hence $L = N$, so N is finitely generated.

(2) $\implies$ (1). Let

$$L_1 \subseteq L_2 \subseteq \cdots$$

be an ascending chain of submodules, and set

$$L := \bigcup_{n \geq 1} L_n.$$

Then L is a submodule of M , so by (2) it is finitely generated, say

$$L = Rx_1 + \cdots + Rx_m.$$

Each generator x_j lies in some L_{n_j} . If $N = \max\{n_1, \dots, n_m\}$, then

$$x_1, \dots, x_m \in L_N,$$

so $L = L_N$. Hence

$$L_n = L_N \quad \text{for all } n \geq N,$$

and the chain stabilizes. □

Proposition 4.5.3. *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be a short exact sequence of R -modules. Then M is Noetherian if and only if both M' and M'' are Noetherian.

Proof. If M is Noetherian, then every submodule of M is finitely generated. Hence the submodule M' is Noetherian. Also every submodule of M'' is the image of a submodule of M , so every submodule of M'' is finitely generated. Thus M'' is Noetherian.

Conversely, assume that M' and M'' are Noetherian. We use the maximal condition. Let $\pi : M \rightarrow M''$ be the quotient map and let $\mathcal{S}$ be a non-empty set of submodules of M . The set

$$\pi(\mathcal{S}) := \{\pi(N) \mid N \in \mathcal{S}\}$$

is a non-empty set of submodules of M'' , so it has a maximal element. Choose $N \in \mathcal{S}$ such that $\pi(N)$ is maximal in $\pi(\mathcal{S})$.

Now consider the set

$$\mathcal{S}_N := \{L \in \mathcal{S} \mid \pi(L) = \pi(N)\}.$$

Then

$$\{L \cap M' \mid L \in \mathcal{S}_N\}$$

is a non-empty set of submodules of M' , hence has a maximal element. Replacing N if necessary, we may assume that $N \cap M'$ is maximal among all $L \cap M'$ with $L \in \mathcal{S}_N$.

We claim that N is maximal in $\mathcal{S}$. Let $N \subseteq L$ with $L \in \mathcal{S}$. Then

$$\pi(N) \subseteq \pi(L).$$

By the maximality of $\pi(N)$, we get $\pi(L) = \pi(N)$, so $L \in \mathcal{S}_N$. Hence

$$N \cap M' \subseteq L \cap M'.$$

By the maximality of $N \cap M'$, it follows that

$$L \cap M' = N \cap M'.$$

For any $x \in L$, since $\pi(x) \in \pi(L) = \pi(N)$, there exists $y \in N$ such that

$$\pi(x) = \pi(y).$$

Then

$$x - y \in \ker \pi = M' \cap L = N \cap M',$$

so $x \in N$. Thus $L = N$, proving that N is maximal. Therefore M is Noetherian. $\square$

Definition 4.5.4. An R -module M is called Artinian if for every descending chain of submodules

$$L_1 \supseteq L_2 \supseteq \cdots$$

there exists n_0 such that

$$L_n = L_{n_0} \quad \text{for all } n \geq n_0.$$

Lemma 4.5.5. An R -module M is Artinian if and only if every non-empty set of submodules of M contains a minimal element.

Proof. The proof is the descending-chain analogue of the Noetherian case. $\square$

Remark 4.5.6. If an R -module M is Noetherian (respectively Artinian), then we also say that M satisfies the ascending chain condition (respectively descending chain condition) on submodules, abbreviated as ACC (respectively DCC).

Proposition 4.5.7. Let

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be a short exact sequence of R -modules. Then M is Artinian if and only if both M' and M'' are Artinian.

Proof. The proof is completely analogous to the Noetherian case, using minimal elements instead of maximal elements. $\square$

Definition 4.5.8. An R -module M is called simple (or irreducible) if it has exactly two submodules:

$$0 \quad \text{and} \quad M.$$

Chapter 5

Homework

5.1 Homework Week 1

Exercise 5.1.1. Let R be a ring, and let M be an R -module. Suppose that S is a ring and

$$\theta : S \rightarrow R$$

is a ring homomorphism satisfying $\theta(1_S) = 1_R$. Let S act on M by

$$sx = \theta(s)x,$$

where $x \in M$ and $s \in S$. Prove that M is an S -module.

Proof. This action is well-defined because $\forall s \in S$, $\theta(s) \in R$ and $\theta(s)x \in M$ for all $x \in M$. We need to verify the module axioms, for all $s, t \in S$ and $x, y \in M$:

- $s(x + y) = \theta(s)(x + y) = \theta(s)x + \theta(s)y = sx + sy$
- $(s + t)x = \theta(s + t)x = \theta(s)x + \theta(t)x = sx + tx$
- $(st)x = \theta(st)x = \theta(s)\theta(t)x = \theta(s)(\theta(t)x) = s(tx)$.
- $1_Sx = \theta(1_S)x = 1_Rx = x$

Hence M is an S -module. □

Exercise 5.1.2. Let R be a ring, and let M and N be two right R -modules. Let $\text{Hom}_{\mathbb{Z}}(M, N)$ be the set of group homomorphisms from M to N (as abelian groups). The action of R on $\text{Hom}_{\mathbb{Z}}(M, N)$ is given as follows: for any $a \in R$ and $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$, define

$$(\varphi a)(x) = \varphi(xa), \quad \forall x \in M.$$

Prove that $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module.

Proof. This action is well-defined. $\forall r \in R$, $\varphi r \in \text{Hom}_{\mathbb{Z}}(M, N)$ for all $\varphi \in \text{Hom}_{\mathbb{Z}}(M, N)$ as $(\varphi r)(x) = \varphi(xr) \in N$ for all $x \in M$. We need to verify the module axioms, for all $r_1, r_2 \in R$ and $\varphi, \psi \in \text{Hom}_{\mathbb{Z}}(M, N)$, $\forall x \in M$:

- $((\varphi + \psi)r)(x) = (\varphi + \psi)(xr) = \varphi(xr) + \psi(xr) = (\varphi r)(x) + (\psi r)(x)$
- $(\varphi(r_1 + r_2))(x) = \varphi(x(r_1 + r_2)) = \varphi(xr_1 + xr_2) = \varphi(xr_1) + \varphi(xr_2) = (\varphi r_1)(x) + (\varphi r_2)(x)$
- $((\varphi r_1)r_2)(x) = (\varphi r_2)(xr_1) = \varphi(xr_2r_1) = (\varphi r_2r_1)(x)$
- $(\varphi 1_R)(x) = \varphi(x 1_R) = \varphi(x)$

Hence $\text{Hom}_{\mathbb{Z}}(M, N)$ is a left R -module. □

Exercise 5.1.3. Let R be a ring, and let M be an R -module. Denote

$$\text{Ann}_R M = \{r \in R \mid rm = 0, \forall m \in M\}.$$

We call M a faithful R -module if $\text{Ann}_R M = \{0\}$. Define the action of the quotient ring $R/\text{Ann}_R M$ on M by

$$\bar{r}x = rx,$$

where $x \in M$, $r \in R$, and $\bar{r} = r + \text{Ann}_R M$. Prove that M is a faithful $R/\text{Ann}_R M$ -module.

Proof. First we show that the action is well-defined. Suppose

$$\bar{r} = \bar{r}' \in R/\text{Ann}_R M.$$

Then $r - r' \in \text{Ann}_R M$, so for every $x \in M$,

$$(r - r')x = 0.$$

Hence $rx = r'x$, so $\bar{r}x$ does not depend on the choice of representative.

The module axioms follow immediately from those of the R -module structure on M .

Now we prove faithfulness. Suppose $\bar{r} \in R/\text{Ann}_R M$ acts trivially on M , that is,

$$\bar{r}x = 0 \quad \forall x \in M.$$

By definition, this means

$$rx = 0 \quad \forall x \in M.$$

Therefore $r \in \text{Ann}_R M$, so $\bar{r} = \bar{0}$ in the quotient ring. Hence

$$\text{Ann}_{R/\text{Ann}_R M}(M) = \{\bar{0}\},$$

and M is faithful as an $R/\text{Ann}_R M$ -module. □

Exercise 5.1.4. Let $\varphi : M \rightarrow M$ be a homomorphism of R -modules such that

$$\varphi\varphi = \varphi.$$

Prove that

$$M = \ker \varphi \oplus \text{im } \varphi.$$

Proof. Let $x \in M$. Then $x = \varphi(x) + (x - \varphi(x))$. Where $\varphi(x) \in \text{im } \varphi$ and $x - \varphi(x) \in \ker \varphi$. Hence $M = \text{im } \varphi + \ker \varphi$. We only need to show that $\ker \varphi \cap \text{im } \varphi = \{0\}$. Suppose $x \in \ker \varphi \cap \text{im } \varphi$. Then $x = \varphi(y)$ for some $y \in M$ and $\varphi(x) = 0$. Then $\varphi(x) = \varphi\varphi(y) = \varphi(y) = x = 0$, so $x = 0$. Hence $\ker \varphi \cap \text{im } \varphi = \{0\}$. $\square$

Exercise 5.1.5. Show that any R -module is a homomorphism image of some free R -module.

Proof. Let M be a left R -module. Consider the free R -module

$$F = \bigoplus_{x \in M} Re_x$$

with basis $\{e_x\}_{x \in M}$ indexed by the underlying set of M . Define an R -module homomorphism

$$\pi : F \rightarrow M$$

by

$$\pi(e_x) = x \quad (x \in M),$$

and extend R -linearly. Thus

$$\pi\left(\sum_{i=1}^n r_i e_{x_i}\right) = \sum_{i=1}^n r_i x_i.$$

This is well-defined and R -linear.

For any $m \in M$, we have

$$\pi(e_m) = m,$$

so π is surjective. Therefore M is a homomorphic image of the free module F . Equivalently,

$$M \cong F / \ker \pi.$$

$\square$

Exercise 5.1.6. Let $\mathbb{Q}$ be the field of rational numbers. Is the additive group $(\mathbb{Q}, +)$ a free $\mathbb{Z}$ -module?

Proof. Claim: any two non-zero element of $\mathbb{Q}$ are linearly dependent over $\mathbb{Z}$. Hence if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

Proof of claim: let $0 \neq x = \frac{a}{b}$ and $0 \neq y = \frac{c}{d}$ be two elements of $\mathbb{Q}$. Then $dcx - aby = 0$. If $a = 0$ or $c = 0$, then $x = 0$ or $y = 0$, which contradicts the assumption. Hence $a, c \neq 0$ which means dc, ab are two non-zero integer coefficients. Hence x and y are linearly dependent over $\mathbb{Z}$. And if $\mathbb{Q}$ is a free $\mathbb{Z}$ -module, then it must have a basis of cardinality 1.

But if $\mathbb{Q}$ is a cyclic $\mathbb{Z}$ -module, then it has only one generator. Suppose it is $z = \frac{e}{f}$. Then $\mathbb{Q} = \{n \frac{e}{f} \mid n \in \mathbb{Z}\}$. But this is not possible because we can easily pick $\frac{1}{2f} \notin \mathbb{Q}$, which is absurd.

Therefore $\mathbb{Q}$ is not a free $\mathbb{Z}$ -module. $\square$

5.2 Homework Week 2

Exercise 5.2.1. Let R be a commutative ring, and suppose that φ is an endomorphism of the free R -module R^m . Prove that if φ is surjective, then φ is injective. Consider that: if φ is injective, is φ surjective?

Proof. Let $\{e_1, \dots, e_m\}$ be the standard basis of R^m , and let $A \in M_m(R)$ be the matrix of φ with respect to this basis. Since φ is surjective, for each j there exists $v_j \in R^m$ with $\varphi(v_j) = e_j$. Write $v_j = \sum_{i=1}^m b_{ij}e_i$ and let $B = (b_{ij}) \in M_m(R)$. Then $AB = I_m$, so

$$\det(A)\det(B) = \det(AB) = 1.$$

Hence $\det(A)$ is a unit in R . It follows that A is invertible (with $A^{-1} = \det(A)^{-1} \operatorname{adj}(A)$), so φ is an isomorphism. In particular φ is injective.

Counterexample for the converse. The converse is false in general. Let $R = \mathbb{Z}$ and $m = 1$, so $R^m = \mathbb{Z}$. Define $\varphi : \mathbb{Z} \rightarrow \mathbb{Z}$ by $\varphi(n) = 2n$. Then φ is injective but not surjective (e.g. $1 \notin \operatorname{im} \varphi$). $\square$

Exercise 5.2.2. Let $\varphi : M \rightarrow N$ be a homomorphism of R -modules. Prove that:

1. φ is a monomorphism if and only if for any R -module T and any two R -module homomorphisms $\xi, \eta : T \rightarrow M$, $\varphi\xi = \varphi\eta$ implies $\xi = \eta$;
2. φ is an epimorphism if and only if for any R -module S and any two R -module homomorphisms $\rho, \theta : N \rightarrow S$, $\rho\varphi = \theta\varphi$ implies $\rho = \theta$.

Proof.

1. $(\Rightarrow)$ Suppose φ is injective and $\varphi\xi = \varphi\eta$ for $\xi, \eta : T \rightarrow M$. Then for every $t \in T$, $\varphi(\xi(t)) = \varphi(\eta(t))$, so $\xi(t) - \eta(t) \in \ker \varphi = \{0\}$. Hence $\xi(t) = \eta(t)$ for all t , i.e. $\xi = \eta$.
 $(\Leftarrow)$ We show $\ker \varphi = \{0\}$. Let $T = \ker \varphi$, $\xi : T \hookrightarrow M$ be the inclusion, and $\eta = 0 : T \rightarrow M$ the zero map. Then $\varphi\xi(t) = \varphi(t) = 0 = \varphi\eta(t)$ for all $t \in T$, so $\varphi\xi = \varphi\eta$. By hypothesis $\xi = \eta$, which means $t = \xi(t) = \eta(t) = 0$ for all $t \in T$. Hence $\ker \varphi = \{0\}$, so φ is injective.
2. $(\Rightarrow)$ Suppose φ is surjective and $\rho\varphi = \theta\varphi$. For any $n \in N$, pick $m \in M$ with $\varphi(m) = n$. Then $\rho(n) = \rho(\varphi(m)) = \theta(\varphi(m)) = \theta(n)$. Hence $\rho = \theta$.
 $(\Leftarrow)$ We show φ is surjective. Let $S = N/\operatorname{im} \varphi$, $\rho : N \rightarrow S$ the canonical projection, and $\theta = 0 : N \rightarrow S$. For any $m \in M$, $\rho(\varphi(m)) = \overline{\varphi(m)} = \bar{0} = \theta(\varphi(m))$, so $\rho\varphi = \theta\varphi$. By hypothesis $\rho = \theta$, i.e. the projection $N \rightarrow N/\operatorname{im} \varphi$ is the zero map. This means $N = \operatorname{im} \varphi$, so φ is surjective.

$\square$

Exercise 5.2.3. Let M and M_i ($1 \leq i \leq n$) be R -modules. Show that $M \simeq \bigoplus_{i=1}^n M_i$ if and only if for every $i \in \{1, \dots, n\}$, there exists an $\varphi_i \in \operatorname{End}_R(M)$ such that $\operatorname{im} \varphi_i \simeq M_i$, $\varphi_i\varphi_j = 0$ for $i \neq j$, $\varphi_i^2 = \varphi_i$, and $\varphi_1 + \varphi_2 + \dots + \varphi_n = \operatorname{id}_M$.

Proof. ($\Rightarrow$) Suppose $f : M \xrightarrow{\sim} \bigoplus_{i=1}^n M_i$. Let $\iota_i : M_i \hookrightarrow \bigoplus_{j=1}^n M_j$ be the canonical inclusion and $\pi_i : \bigoplus_{j=1}^n M_j \twoheadrightarrow M_i$ the canonical projection. Define

$$\varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f \in \text{End}_R(M).$$

Then:

- $\varphi_i^2 = f^{-1} \iota_i \pi_i f f^{-1} \iota_i \pi_i f = f^{-1} \iota_i (\pi_i \iota_i) \pi_i f = f^{-1} \iota_i \pi_i f = \varphi_i$, since $\pi_i \iota_i = \text{id}_{M_i}$.
- For $i \neq j$: $\varphi_i \varphi_j = f^{-1} \iota_i (\pi_i \iota_j) \pi_j f = 0$, since $\pi_i \iota_j = 0$ when $i \neq j$.
- $\sum_{i=1}^n \varphi_i = f^{-1} (\sum_{i=1}^n \iota_i \pi_i) f = f^{-1} \circ \text{id} \circ f = \text{id}_M$, since $\sum_i \iota_i \pi_i = \text{id}_{\bigoplus M_j}$.
- $\text{im } \varphi_i = f^{-1} \circ \iota_i \circ \pi_i \circ f(M) \simeq M_i$.

($\Leftarrow$) Suppose such φ_i exist. Define $g : M \rightarrow \bigoplus_{i=1}^n \text{im } \varphi_i$ by $g(x) = (\varphi_1(x), \dots, \varphi_n(x))$.

Surjective: for any $(y_1, \dots, y_n) \in \bigoplus \text{im } \varphi_i$, let $x = y_1 + \dots + y_n \in M$. Then

$$\varphi_j(x) = \sum_{i=1}^n \varphi_j(y_i) = \sum_{i=1}^n \varphi_j(\varphi_i(x_i)) = \varphi_j(\varphi_j(x_j)) = \varphi_j^2(x_j) = \varphi_j(x_j) = y_j,$$

where we used $\varphi_j \varphi_i = 0$ for $j \neq i$ and $\varphi_j^2 = \varphi_j$. Hence $g(x) = (y_1, \dots, y_n)$.

Injective: if $g(x) = 0$, then $\varphi_i(x) = 0$ for all i , so $x = \text{id}_M(x) = \sum_i \varphi_i(x) = 0$.

Since $\text{im } \varphi_i \simeq M_i$, we conclude $M \simeq \bigoplus_{i=1}^n M_i$. □

Exercise 5.2.4.

1. Prove $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \mathbb{Z}/(m, n)\mathbb{Z}$, where $m, n \in \mathbb{Z}$.
2. Let A be an abelian group, prove $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.
3. Let A, B be two finitely generated abelian groups. What is $A \otimes_{\mathbb{Z}} B$?

Proof.

1. We use the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$. Tensoring with $\mathbb{Z}/m\mathbb{Z}$ over $\mathbb{Z}$ gives the right-exact sequence

$$\mathbb{Z}/m\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z}/m\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow 0.$$

Hence $\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq (\mathbb{Z}/m\mathbb{Z})/n(\mathbb{Z}/m\mathbb{Z})$. Now $n(\mathbb{Z}/m\mathbb{Z}) = (n\mathbb{Z} + m\mathbb{Z})/m\mathbb{Z}$. Since $n\mathbb{Z} + m\mathbb{Z} = (m, n)\mathbb{Z}$, we get

$$\mathbb{Z}/m\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq \frac{\mathbb{Z}/m\mathbb{Z}}{(m, n)\mathbb{Z}/m\mathbb{Z}} \simeq \mathbb{Z}/(m, n)\mathbb{Z}.$$

2. Similarly, tensor the right-exact sequence $\mathbb{Z} \xrightarrow{\cdot n} \mathbb{Z} \rightarrow \mathbb{Z}/n\mathbb{Z} \rightarrow 0$ with A :

$$A \xrightarrow{\cdot n} A \rightarrow A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \rightarrow 0.$$

By right-exactness, $A \otimes_{\mathbb{Z}} \mathbb{Z}/n\mathbb{Z} \simeq A/nA$.

3. By the fundamental theorem of finitely generated abelian groups,

$$A \simeq \mathbb{Z}^r \oplus \bigoplus_{i=1}^s \mathbb{Z}/d_i\mathbb{Z}, \quad B \simeq \mathbb{Z}^t \oplus \bigoplus_{j=1}^u \mathbb{Z}/e_j\mathbb{Z}.$$

Since tensor product distributes over direct sums, using (1), (2), and $\mathbb{Z} \otimes_{\mathbb{Z}} M \simeq M$, we get

$$A \otimes_{\mathbb{Z}} B \simeq \mathbb{Z}^{rt} \oplus \bigoplus_{j=1}^u (\mathbb{Z}/e_j\mathbb{Z})^r \oplus \bigoplus_{i=1}^s (\mathbb{Z}/d_i\mathbb{Z})^t \oplus \bigoplus_{i=1}^s \bigoplus_{j=1}^u \mathbb{Z}/(d_i, e_j)\mathbb{Z}.$$

□

Exercise 5.2.5. Let $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ be the unique monomorphism. Prove that

$$\text{id} \otimes \varphi : \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z}$$

is a zero homomorphism.

Proof. The unique monomorphism $\varphi : \mathbb{Z}/2\mathbb{Z} \rightarrow \mathbb{Z}/4\mathbb{Z}$ is given by $\varphi(\bar{1}) = \bar{2}$. By the previous exercise, $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/4\mathbb{Z} \simeq \mathbb{Z}/2\mathbb{Z}$.

It suffices to check on the generator $\bar{1} \otimes \bar{1}$ of $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$:

$$(\text{id} \otimes \varphi)(\bar{1} \otimes \bar{1}) = \bar{1} \otimes \varphi(\bar{1}) = \bar{1} \otimes \bar{2} = \bar{1} \otimes 2 \cdot \bar{1} = 2 \cdot (\bar{1} \otimes \bar{1}) = (2 \cdot \bar{1}) \otimes \bar{1} = \bar{0} \otimes \bar{1} = 0,$$

where we used $2 \cdot \bar{1} = \bar{0}$ in $\mathbb{Z}/2\mathbb{Z}$. Since the generator maps to 0, the map $\text{id} \otimes \varphi$ is the zero homomorphism.

This also shows that tensoring a monomorphism need not yield a monomorphism, i.e. the functor $\mathbb{Z}/2\mathbb{Z} \otimes_{\mathbb{Z}} -$ is not left-exact. □

Exercise 5.2.6. Let V_1 and V_2 be two vector spaces over a field F with dimension n and m respectively. Let $\{\epsilon_1, \dots, \epsilon_n\}$ and $\{\eta_1, \dots, \eta_m\}$ be bases of V_1 and V_2 respectively. Suppose that $\mathcal{A}$ (resp. $\mathcal{B}$) is an F -linear map of V_1 (resp. V_2), and it is represented by the matrix $A = (a_{ij})_{n \times n}$ (resp. $B = (b_{ij})_{m \times m}$) with respect to the above bases. Prove that

$$\{\epsilon_i \otimes \eta_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$$

is an F -basis of $V_1 \otimes_F V_2$. Determine the matrix representing the F -linear map $\mathcal{A} \otimes \mathcal{B}$ under this basis.

Proof. Step 1: $\{\epsilon_i \otimes \eta_j\}$ is a basis.

Spanning. Every element of $V_1 \otimes_F V_2$ is a finite sum of simple tensors $v \otimes w$. Writing $v = \sum_i a_i \epsilon_i$ and $w = \sum_j b_j \eta_j$, by bilinearity of the tensor product we get

$$v \otimes w = \sum_{i,j} a_i b_j (\epsilon_i \otimes \eta_j).$$

Hence $\{\epsilon_i \otimes \eta_j\}$ spans $V_1 \otimes_F V_2$.

Linear independence. Suppose $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$. For each pair (p, q) , let $f_p \in V_1^*$ and $g_q \in V_2^*$ be the dual basis functionals: $f_p(\epsilon_i) = \delta_{pi}$ and $g_q(\eta_j) = \delta_{qj}$. The bilinear map $(v, w) \mapsto f_p(v)g_q(w)$ from $V_1 \times V_2$ to F induces (by the universal property of the tensor product) an F -linear map $\Phi_{pq} : V_1 \otimes_F V_2 \rightarrow F$ with $\Phi_{pq}(\epsilon_i \otimes \eta_j) = \delta_{pi}\delta_{qj}$. Applying Φ_{pq} to the relation $\sum_{i,j} c_{ij}(\epsilon_i \otimes \eta_j) = 0$ yields $c_{pq} = 0$. Since (p, q) was arbitrary, all $c_{ij} = 0$.

Therefore $\{\epsilon_i \otimes \eta_j : 1 \leq i \leq n, 1 \leq j \leq m\}$ is an F -basis of $V_1 \otimes_F V_2$, and $\dim_F(V_1 \otimes_F V_2) = nm$.

Step 2: matrix of $\mathcal{A} \otimes \mathcal{B}$.

Order the basis lexicographically: $\epsilon_1 \otimes \eta_1, \epsilon_1 \otimes \eta_2, \dots, \epsilon_1 \otimes \eta_m, \epsilon_2 \otimes \eta_1, \dots, \epsilon_n \otimes \eta_m$. Since $\mathcal{A}(\epsilon_i) = \sum_{k=1}^n a_{ki}\epsilon_k$ and $\mathcal{B}(\eta_j) = \sum_{l=1}^m b_{lj}\eta_l$, we have

$$(\mathcal{A} \otimes \mathcal{B})(\epsilon_i \otimes \eta_j) = \mathcal{A}(\epsilon_i) \otimes \mathcal{B}(\eta_j) = \sum_{k=1}^n \sum_{l=1}^m a_{ki}b_{lj}(\epsilon_k \otimes \eta_l).$$

The entry in row (k, l) and column (i, j) of the representing matrix is $a_{ki}b_{lj}$. Written as an $nm \times nm$ block matrix, this is the **Kronecker product**

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B & \cdots & a_{1n}B \\ a_{21}B & a_{22}B & \cdots & a_{2n}B \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1}B & a_{n2}B & \cdots & a_{nn}B \end{pmatrix}.$$

□

5.3 Homework Week 3

Exercise 5.3.1. Let M, N, M_i ($i \in I$) and N_j ($j \in J$) be R -modules. Prove the following isomorphisms of abelian groups:

1.

$$\operatorname{Hom}_R\left(\bigoplus_{i \in I} M_i, N\right) \cong \prod_{i \in I} \operatorname{Hom}_R(M_i, N).$$

2.

$$\operatorname{Hom}_R\left(M, \prod_{j \in J} N_j\right) \cong \prod_{j \in J} \operatorname{Hom}_R(M, N_j).$$

Proof. We prove the two statements separately.

(1) Define

$$\Phi : \operatorname{Hom}_R\left(\bigoplus_{i \in I} M_i, N\right) \longrightarrow \prod_{i \in I} \operatorname{Hom}_R(M_i, N)$$

by

$$\Phi(f) = (f \circ \iota_i)_{i \in I},$$

where

$$\iota_i : M_i \rightarrow \bigoplus_{i \in I} M_i$$

is the canonical inclusion.

We show that Φ is an isomorphism.

First, Φ is a group homomorphism, since composition is compatible with addition:

$$\Phi(f + g) = ((f + g) \circ \iota_i)_{i \in I} = (f \circ \iota_i + g \circ \iota_i)_{i \in I} = \Phi(f) + \Phi(g).$$

Now let $(f_i)_{i \in I} \in \prod_{i \in I} \text{Hom}_R(M_i, N)$. Define

$$f : \bigoplus_{i \in I} M_i \rightarrow N$$

by

$$f((m_i)_{i \in I}) = \sum_{i \in I} f_i(m_i).$$

This is well-defined, because an element of $\bigoplus_{i \in I} M_i$ has only finitely many nonzero components, so the above sum is finite. It is immediate that f is R -linear, and

$$f \circ \iota_i = f_i \quad \text{for all } i \in I.$$

Hence $\Phi(f) = (f_i)_{i \in I}$, so Φ is surjective.

To prove injectivity, suppose $\Phi(f) = \Phi(g)$. Then

$$f \circ \iota_i = g \circ \iota_i \quad \text{for all } i \in I.$$

Take any $x = (m_i)_{i \in I} \in \bigoplus_{i \in I} M_i$. Since x has finite support, we may write

$$x = \sum_{i \in I} \iota_i(m_i),$$

with only finitely many nonzero terms. Therefore,

$$f(x) = \sum_{i \in I} f(\iota_i(m_i)) = \sum_{i \in I} g(\iota_i(m_i)) = g(x).$$

So $f = g$, and Φ is injective.

Thus Φ is an isomorphism:

$$\text{Hom}_R\left(\bigoplus_{i \in I} M_i, N\right) \cong \prod_{i \in I} \text{Hom}_R(M_i, N).$$

(2) Let

$$\pi_j : \prod_{j \in J} N_j \rightarrow N_j$$

be the canonical projection. Define

$$\Psi : \text{Hom}_R \left(M, \prod_{j \in J} N_j \right) \longrightarrow \prod_{j \in J} \text{Hom}_R(M, N_j)$$

by

$$\Psi(f) = (\pi_j \circ f)_{j \in J}.$$

Again, Ψ is a group homomorphism.

We show that Ψ is an isomorphism. Given a family

$$(g_j)_{j \in J} \in \prod_{j \in J} \text{Hom}_R(M, N_j),$$

define

$$g : M \rightarrow \prod_{j \in J} N_j$$

by

$$g(m) = (g_j(m))_{j \in J}.$$

Since each g_j is R -linear, g is also R -linear. Moreover,

$$\pi_j \circ g = g_j \quad \text{for all } j \in J,$$

so

$$\Psi(g) = (g_j)_{j \in J}.$$

Hence Ψ is surjective.

For injectivity, suppose $\Psi(f) = \Psi(g)$. Then

$$\pi_j \circ f = \pi_j \circ g \quad \text{for all } j \in J.$$

Thus for every $m \in M$ and every $j \in J$,

$$\pi_j(f(m)) = \pi_j(g(m)).$$

Since two elements of $\prod_{j \in J} N_j$ are equal if and only if all their components are equal, it follows that

$$f(m) = g(m) \quad \text{for all } m \in M.$$

Hence $f = g$, and Ψ is injective.

Therefore

$$\text{Hom}_R \left(M, \prod_{j \in J} N_j \right) \cong \prod_{j \in J} \text{Hom}_R(M, N_j).$$

This completes the proof. $\square$

Exercise 5.3.2 (Five Lemma). *Suppose that the following diagram of homomorphisms of*

R-modules

$$\begin{array}{ccccccccc}
 M_1 & \longrightarrow & M_2 & \longrightarrow & M_3 & \longrightarrow & M_4 & \longrightarrow & M_5 \\
 \varphi_1 \downarrow & & \varphi_2 \downarrow & & \varphi_3 \downarrow & & \varphi_4 \downarrow & & \varphi_5 \downarrow \\
 N_1 & \longrightarrow & N_2 & \longrightarrow & N_3 & \longrightarrow & N_4 & \longrightarrow & N_5
 \end{array}$$

is commutative and has exact rows. Prove that:

1. If φ_1 is surjective, and φ_2 and φ_4 are injective, then φ_3 is injective.
2. If φ_5 is injective, and φ_2 and φ_4 are surjective, then φ_3 is surjective.

Proof. We prove the two assertions separately.

(1) Assume that φ_1 is surjective and that φ_2, φ_4 are injective. We show that φ_3 is injective.

Let $x \in M_3$ and suppose that

$$\varphi_3(x) = 0.$$

We must prove that $x = 0$.

Let $\alpha_3 : M_3 \rightarrow M_4$ and $\beta_3 : N_3 \rightarrow N_4$ denote the third horizontal maps. By commutativity,

$$\varphi_4(\alpha_3(x)) = \beta_3(\varphi_3(x)) = \beta_3(0) = 0.$$

Since φ_4 is injective, it follows that $\alpha_3(x) = 0$. Exactness of the top row at M_3 yields

$$x = \alpha_2(y)$$

for some $y \in M_2$, where $\alpha_2 : M_2 \rightarrow M_3$ is the second horizontal map.

Again by commutativity,

$$0 = \varphi_3(x) = \varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)),$$

where $\beta_2 : N_2 \rightarrow N_3$ is the second horizontal map. Hence $\varphi_2(y) \in \ker(\beta_2)$. By exactness of the bottom row at N_2 ,

$$\ker(\beta_2) = \operatorname{im}(\beta_1),$$

so there exists $z \in N_1$ such that

$$\beta_1(z) = \varphi_2(y),$$

where $\beta_1 : N_1 \rightarrow N_2$ is the first horizontal map.

Since φ_1 is surjective, there exists $w \in M_1$ such that

$$\varphi_1(w) = z.$$

Using commutativity once more,

$$\varphi_2(\alpha_1(w)) = \beta_1(\varphi_1(w)) = \beta_1(z) = \varphi_2(y),$$

where $\alpha_1 : M_1 \rightarrow M_2$ is the first horizontal map. Because φ_2 is injective, we get

$$\alpha_1(w) = y.$$

Therefore

$$x = \alpha_2(y) = \alpha_2(\alpha_1(w)) = 0,$$

since the composition of two consecutive maps in an exact sequence is zero. Thus $\ker(\varphi_3) = 0$, so φ_3 is injective.

(2) Assume that φ_5 is injective and that φ_2, φ_4 are surjective. We show that φ_3 is surjective.

Let $n \in N_3$. We must find $m \in M_3$ such that

$$\varphi_3(m) = n.$$

Let $\beta_3 : N_3 \rightarrow N_4$ and $\alpha_3 : M_3 \rightarrow M_4$ denote the third horizontal maps. Since φ_4 is surjective, there exists $u \in M_4$ such that

$$\varphi_4(u) = \beta_3(n).$$

Let $\alpha_4 : M_4 \rightarrow M_5$ and $\beta_4 : N_4 \rightarrow N_5$ be the next horizontal maps. Then

$$\varphi_5(\alpha_4(u)) = \beta_4(\varphi_4(u)) = \beta_4(\beta_3(n)) = 0.$$

Because φ_5 is injective, it follows that $\alpha_4(u) = 0$. Exactness of the top row at M_4 shows that there exists $m \in M_3$ such that

$$\alpha_3(m) = u.$$

Consider

$$n - \varphi_3(m) \in N_3.$$

By commutativity,

$$\beta_3(n - \varphi_3(m)) = \beta_3(n) - \beta_3(\varphi_3(m)) = \beta_3(n) - \varphi_4(\alpha_3(m)) = \beta_3(n) - \varphi_4(u) = 0.$$

Hence $n - \varphi_3(m) \in \ker(\beta_3)$. By exactness of the bottom row at N_3 ,

$$\ker(\beta_3) = \text{im}(\beta_2),$$

so there exists $v \in N_2$ such that

$$\beta_2(v) = n - \varphi_3(m),$$

where $\beta_2 : N_2 \rightarrow N_3$ is the second horizontal map.

Since φ_2 is surjective, there exists $y \in M_2$ such that

$$\varphi_2(y) = v.$$

Then

$$\varphi_3(\alpha_2(y)) = \beta_2(\varphi_2(y)) = \beta_2(v) = n - \varphi_3(m),$$

where $\alpha_2 : M_2 \rightarrow M_3$ is the second horizontal map. Therefore

$$n = \varphi_3(m) + \varphi_3(\alpha_2(y)) = \varphi_3(m + \alpha_2(y)).$$

Thus n lies in the image of φ_3 , proving that φ_3 is surjective. $\square$

Exercise 5.3.3 (Snake Lemma). *Suppose that the following diagram of homomorphisms of R -modules*

$$\begin{array}{ccccccccc} 0 & \longrightarrow & M' & \xrightarrow{\alpha} & M & \xrightarrow{\beta} & M'' & \longrightarrow & 0 \\ & & \downarrow \varphi' & & \downarrow \varphi & & \downarrow \varphi'' & & \\ 0 & \longrightarrow & N' & \xrightarrow{\mu} & N & \xrightarrow{\nu} & N'' & \longrightarrow & 0 \end{array}$$

is commutative and has exact rows. Prove that

$$0 \longrightarrow \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \operatorname{coker} \varphi' \xrightarrow{\mu'} \operatorname{coker} \varphi \xrightarrow{\nu'} \operatorname{coker} \varphi'' \longrightarrow 0$$

is a long exact sequence, where $\alpha', \beta', \mu', \nu'$ are the homomorphisms induced by α, β, μ, ν , respectively.

Proof. Since the rows are exact, α and μ are injective, while β and ν are surjective.

We first define the connecting homomorphism

$$\delta : \ker \varphi'' \rightarrow \operatorname{coker} \varphi'.$$

Let $x \in \ker \varphi'' \subseteq M''$. Since $\beta : M \rightarrow M''$ is surjective, choose $y \in M$ such that

$$\beta(y) = x.$$

By commutativity,

$$\nu(\varphi(y)) = \varphi''(\beta(y)) = \varphi''(x) = 0.$$

Hence $\varphi(y) \in \ker \nu = \operatorname{im} \mu$. Therefore there exists $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Define

$$\delta(x) = \bar{z} = z + \operatorname{im} \varphi' \in \operatorname{coker} \varphi'.$$

We must check that this is well-defined. First, suppose $y_1, y_2 \in M$ both satisfy

$$\beta(y_1) = \beta(y_2) = x.$$

Then $y_1 - y_2 \in \ker \beta = \operatorname{im} \alpha$, so there exists $m' \in M'$ such that

$$y_1 - y_2 = \alpha(m').$$

If $\mu(z_i) = \varphi(y_i)$ for $i = 1, 2$, then

$$\mu(z_1 - z_2) = \varphi(y_1) - \varphi(y_2) = \varphi(\alpha(m')) = \mu(\varphi'(m')),$$

where we used commutativity. Since μ is injective, it follows that

$$z_1 - z_2 = \varphi'(m') \in \text{im } \varphi'.$$

Thus $\overline{z_1} = \overline{z_2}$ in $\text{coker } \varphi'$, so $\delta(x)$ is independent of the choice of y .

Next, if $z, z' \in N'$ both satisfy

$$\mu(z) = \mu(z') = \varphi(y),$$

then $\mu(z - z') = 0$, and since μ is injective, $z = z'$. Hence $\delta(x)$ is also independent of the choice of z .

It is straightforward to verify that δ is a homomorphism.

Now we prove exactness.

Exactness at $\ker \varphi'$. Since α induces α' , we have

$$\alpha'(\ker \varphi') \subseteq \ker \varphi.$$

Because α is injective, so is α' . Hence

$$\ker \alpha' = 0.$$

Therefore the sequence is exact at $\ker \varphi'$.

Exactness at $\ker \varphi$. We show that

$$\text{im } \alpha' = \ker \beta'.$$

Take $x' \in \ker \varphi'$. Then

$$\beta'(\alpha'(x')) = \beta(\alpha(x')) = 0,$$

so $\text{im } \alpha' \subseteq \ker \beta'$.

Conversely, let $y \in \ker \varphi$ satisfy $\beta'(y) = 0$. Then $\beta(y) = 0$, hence by exactness of the top row there exists $x' \in M'$ such that

$$\alpha(x') = y.$$

Since $\varphi(y) = 0$, we get

$$0 = \varphi(\alpha(x')) = \mu(\varphi'(x')).$$

As μ is injective, $\varphi'(x') = 0$, so $x' \in \ker \varphi'$. Thus

$$y = \alpha'(x'),$$

proving $\ker \beta' \subseteq \operatorname{im} \alpha'$.

Exactness at $\ker \varphi''$. We show that

$$\operatorname{im} \beta' = \ker \delta.$$

First let $y \in \ker \varphi$. Then $\beta(y) \in \ker \varphi''$, since

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = 0.$$

Thus $\beta'(y) \in \ker \varphi''$. To compute $\delta(\beta'(y))$, choose the lift y itself. Then $\varphi(y) = 0$, so we may take $z = 0$, and hence

$$\delta(\beta'(y)) = \bar{0} = 0.$$

Therefore $\operatorname{im} \beta' \subseteq \ker \delta$.

Conversely, let $x \in \ker \varphi''$ and suppose $\delta(x) = 0$. Choose $y \in M$ with

$$\beta(y) = x,$$

and choose $z \in N'$ such that

$$\mu(z) = \varphi(y).$$

Since $\delta(x) = \bar{z} = 0$ in $\operatorname{coker} \varphi'$, there exists $x' \in M'$ such that

$$z = \varphi'(x').$$

Hence

$$\varphi(y - \alpha(x')) = \varphi(y) - \varphi(\alpha(x')) = \mu(z) - \mu(\varphi'(x')) = 0.$$

So $y - \alpha(x') \in \ker \varphi$, and

$$\beta'(y - \alpha(x')) = \beta(y - \alpha(x')) = \beta(y) = x,$$

because $\beta\alpha = 0$. Thus $x \in \operatorname{im} \beta'$, proving

$$\ker \delta \subseteq \operatorname{im} \beta'.$$

Exactness at $\operatorname{coker} \varphi'$. We show that

$$\operatorname{im} \delta = \ker \mu'.$$

Let $x \in \ker \varphi''$, and write

$$\delta(x) = \bar{z} \in \operatorname{coker} \varphi'$$

as above, with $\mu(z) = \varphi(y)$. Then

$$\mu'(\delta(x)) = \overline{\mu(z)} = \overline{\varphi(y)} = 0$$

in $\text{coker } \varphi$. Hence $\text{im } \delta \subseteq \ker \mu'$.

Conversely, let $\bar{z} \in \text{coker } \varphi'$ satisfy

$$\mu'(\bar{z}) = 0.$$

This means that $\mu(z) \in \text{im } \varphi$, so there exists $y \in M$ such that

$$\varphi(y) = \mu(z).$$

Applying ν , we get

$$\varphi''(\beta(y)) = \nu(\varphi(y)) = \nu(\mu(z)) = 0.$$

Thus $\beta(y) \in \ker \varphi''$. By construction of δ ,

$$\delta(\beta(y)) = \bar{z}.$$

So $\bar{z} \in \text{im } \delta$, proving

$$\ker \mu' \subseteq \text{im } \delta.$$

Exactness at $\text{coker } \varphi$. We show that

$$\text{im } \mu' = \ker \nu'.$$

For $\bar{z} \in \text{coker } \varphi'$, we have

$$\nu'(\mu'(\bar{z})) = \overline{\nu(\mu(z))} = \bar{0} = 0,$$

so $\text{im } \mu' \subseteq \ker \nu'$.

Conversely, let $\bar{n} \in \text{coker } \varphi$ satisfy

$$\nu'(\bar{n}) = 0.$$

Then $\overline{\nu(n)} = 0$ in $\text{coker } \varphi''$, so there exists $m'' \in M''$ such that

$$\varphi''(m'') = \nu(n).$$

Since β is surjective, choose $m \in M$ such that

$$\beta(m) = m''.$$

Then

$$\nu(n - \varphi(m)) = \nu(n) - \varphi''(\beta(m)) = \nu(n) - \varphi''(m'') = 0.$$

Hence

$$n - \varphi(m) \in \ker \nu = \text{im } \mu,$$

so there exists $z \in N'$ such that

$$\mu(z) = n - \varphi(m).$$

Therefore in $\text{coker } \varphi$,

$$\bar{n} = \overline{\mu(z)} = \mu'(\bar{z}),$$

showing that $\bar{n} \in \text{im } \mu'$. Thus

$$\ker \nu' \subseteq \text{im } \mu'.$$

Exactness at $\text{coker } \varphi''$ and termination. Since $\nu : N \rightarrow N''$ is surjective, the induced map

$$\nu' : \text{coker } \varphi \rightarrow \text{coker } \varphi''$$

is surjective as well: for any $\bar{n}'' \in \text{coker } \varphi''$, choose $n \in N$ such that

$$\nu(n) = n'',$$

then

$$\nu'(\bar{n}) = \bar{n}''.$$

Hence

$$\text{im } \nu' = \text{coker } \varphi'',$$

which is equivalent to exactness at $\text{coker } \varphi''$ and the final 0.

Collecting all the above, we obtain the exact sequence

$$0 \longrightarrow \ker \varphi' \xrightarrow{\alpha'} \ker \varphi \xrightarrow{\beta'} \ker \varphi'' \xrightarrow{\delta} \text{coker } \varphi' \xrightarrow{\mu'} \text{coker } \varphi \xrightarrow{\nu'} \text{coker } \varphi'' \longrightarrow 0.$$

□

5.4 Homework Week 4

Exercise 5.4.1. Let J be an R -module. Show that the following are equivalent.

1. J is an injective module.
2. For any R -module monomorphism $\varphi : J \rightarrow M$, there exists an R -module homomorphism $\psi : M \rightarrow J$ such that $\psi\varphi = \text{id}_J$.
3. If J is a submodule of some R -module M , then J is a direct summand of M .

Proof. We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

$(1) \Rightarrow (2)$. Assume that J is injective, and let $\varphi : J \rightarrow M$ be a monomorphism. Apply the defining property of injectivity to the monomorphism φ and the map $\text{id}_J : J \rightarrow J$:

$$\begin{array}{ccc} J & \xhookrightarrow{\varphi} & M \\ \text{id}_J \downarrow & \swarrow \exists \psi & \\ J & & \end{array}$$

Then there exists an R -module homomorphism $\psi : M \rightarrow J$ such that

$$\psi \circ \varphi = \text{id}_J.$$

(2) $\Rightarrow$ (3). Assume (2). If $J \subseteq M$, let $\iota: J \hookrightarrow M$ be the inclusion map. By (2), there exists $\psi: M \rightarrow J$ such that

$$\psi \circ \iota = \text{id}_J.$$

We claim that

$$M = J \oplus \ker \psi.$$

Indeed, for any $m \in M$,

$$m = \iota(\psi(m)) + (m - \iota(\psi(m))),$$

and

$$\psi(m - \iota(\psi(m))) = \psi(m) - \psi(\iota(\psi(m))) = \psi(m) - \psi(m) = 0,$$

so $m - \iota(\psi(m)) \in \ker \psi$. Hence $M = J + \ker \psi$. If $x \in J \cap \ker \psi$, then

$$x = (\psi \circ \iota)(x) = \psi(x) = 0,$$

so $J \cap \ker \psi = 0$. Therefore $M = J \oplus \ker \psi$, and J is a direct summand of M .

(3) $\Rightarrow$ (1). Assume (3). To prove that J is injective, let

$$i: A \hookrightarrow B$$

be a monomorphism of R -modules, and let

$$f: A \rightarrow J$$

be an R -module homomorphism. We must show that f extends to a map $g: B \rightarrow J$.

Consider the quotient module

$$P = (B \oplus J)/N, \quad N = \{(i(a), -f(a)) \mid a \in A\}.$$

Define maps

$$\eta: J \rightarrow P, \quad \eta(j) = [(0, j)],$$

and

$$\beta: B \rightarrow P, \quad \beta(b) = [(b, 0)].$$

We first show that η is injective. If $\eta(j) = 0$, then $(0, j) \in N$, so for some $a \in A$,

$$(0, j) = (i(a), -f(a)).$$

Hence $i(a) = 0$. Since i is injective, $a = 0$, and therefore $j = -f(0) = 0$. Thus η is a monomorphism, so we may identify J with the submodule $\eta(J) \subseteq P$.

By (3), this submodule is a direct summand of P . Therefore there exists an R -module homomorphism

$$s: P \rightarrow J$$

such that

$$s \circ \eta = \text{id}_J.$$

Now define

$$g = s \circ \beta: B \rightarrow J.$$

For any $a \in A$, we have in P :

$$[(i(a), 0)] = [(0, f(a))],$$

because

$$(i(a), 0) - (0, f(a)) = (i(a), -f(a)) \in N.$$

Hence

$$g(i(a)) = s([(i(a), 0)]) = s([(0, f(a))]) = (s \circ \eta)(f(a)) = f(a).$$

So $g \circ i = f$, and f extends to B . Therefore J is injective. $\square$

Exercise 5.4.2. Suppose that M_i ($i \in I$) are R -modules. Prove that

$$\bigoplus_{i \in I} M_i$$

is a flat R -module if and only if each M_i is a flat R -module.

Proof. Set

$$M = \bigoplus_{i \in I} M_i.$$

First assume that each M_i is flat. Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R -modules. Since each M_i is flat, for every $i \in I$ the sequence

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Taking direct sums over $i \in I$, we obtain an exact sequence

$$0 \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R A) \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R B) \longrightarrow \bigoplus_{i \in I} (M_i \otimes_R C) \longrightarrow 0.$$

Using the canonical isomorphisms

$$\left(\bigoplus_{i \in I} M_i \right) \otimes_R X \cong \bigoplus_{i \in I} (M_i \otimes_R X) \quad (X = A, B, C),$$

we get

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0.$$

Hence M is flat.

Conversely, assume that M is flat. Fix $i \in I$, and write

$$M = M_i \oplus N_i, \quad N_i = \bigoplus_{j \neq i} M_j.$$

Let

$$0 \longrightarrow A \longrightarrow B \longrightarrow C \longrightarrow 0$$

be a short exact sequence of R -modules. Since M is flat, the sequence

$$0 \longrightarrow M \otimes_R A \longrightarrow M \otimes_R B \longrightarrow M \otimes_R C \longrightarrow 0$$

is exact. Using

$$M \otimes_R X \cong (M_i \otimes_R X) \oplus (N_i \otimes_R X) \quad (X = A, B, C),$$

this becomes

$$0 \longrightarrow (M_i \otimes_R A) \oplus (N_i \otimes_R A) \longrightarrow (M_i \otimes_R B) \oplus (N_i \otimes_R B) \longrightarrow (M_i \otimes_R C) \oplus (N_i \otimes_R C) \longrightarrow 0.$$

Since kernels and images of direct sum maps decompose componentwise, it follows that

$$0 \longrightarrow M_i \otimes_R A \longrightarrow M_i \otimes_R B \longrightarrow M_i \otimes_R C \longrightarrow 0$$

is exact. Therefore M_i is flat. As i was arbitrary, each M_i is flat. $\square$

Exercise 5.4.3. *Prove that the following three conditions on an R -module V are equivalent.*

1. (Ascending chain condition.) *For every infinite chain of R -submodules of V ,*

$$V_1 \subseteq V_2 \subseteq \cdots \subseteq V_n \subseteq \cdots,$$

there is a number m such that

$$V_m = V_{m+1} = \cdots.$$

2. *Any non-empty set of submodules of V contains a maximal element with respect to inclusion.*

3. *Every R -submodule of V is finitely generated.*

Proof. We prove $(1) \Rightarrow (2) \Rightarrow (3) \Rightarrow (1)$.

$(1) \Rightarrow (2)$. Assume that (2) fails. Then there exists a non-empty set $\mathcal{S}$ of submodules of V having no maximal element. Choose $V_1 \in \mathcal{S}$. Since V_1 is not maximal in $\mathcal{S}$, there exists $V_2 \in \mathcal{S}$ such that

$$V_1 \subsetneq V_2.$$

Inductively, having chosen $V_n \in \mathcal{S}$, since V_n is not maximal there exists $V_{n+1} \in \mathcal{S}$ with

$$V_n \subsetneq V_{n+1}.$$

Thus we obtain a strictly ascending chain

$$V_1 \subsetneq V_2 \subsetneq \cdots,$$

which contradicts (1). Hence every non-empty set of submodules contains a maximal element.

(2) $\Rightarrow$ (3). Let U be a submodule of V . Consider the set

$$\mathcal{T} = \{N \leq U \mid N \text{ is finitely generated}\}.$$

This set is non-empty, since $0 \in \mathcal{T}$. By (2), $\mathcal{T}$ has a maximal element, say N .

We claim that $N = U$. If not, choose $u \in U \setminus N$. Then

$$N + Ru$$

is a finitely generated submodule of U properly containing N , contradicting the maximality of N in $\mathcal{T}$. Therefore $U = N$, so U is finitely generated. Hence every submodule of V is finitely generated.

(3) $\Rightarrow$ (1). Let

$$V_1 \subseteq V_2 \subseteq \cdots$$

be an ascending chain of submodules of V . Set

$$U = \bigcup_{n \geq 1} V_n.$$

Since the chain is ascending, U is a submodule of V . By (3), U is finitely generated, say

$$U = Ru_1 + \cdots + Ru_r.$$

For each j , choose n_j such that $u_j \in V_{n_j}$, and let

$$m = \max\{n_1, \dots, n_r\}.$$

Then $u_1, \dots, u_r \in V_m$, so $U \subseteq V_m$. Since always $V_m \subseteq U$, we get $U = V_m$. Therefore for all $n \geq m$,

$$V_n = U = V_m,$$

so the chain stabilizes. Hence (1) holds. $\square$

Exercise 5.4.4. Let W be an R -submodule of V . Prove that V is Noetherian if and only if both W and V/W are Noetherian.

Proof. We use the characterization that an R -module is Noetherian if and only if every submodule is finitely generated.

Assume first that V is Noetherian. Then every submodule of V is finitely generated.

Since every submodule of W is also a submodule of V , it follows that every submodule of W is finitely generated. Hence W is Noetherian.

Now let $\overline{U}$ be a submodule of V/W , and let

$$\pi: V \rightarrow V/W$$

be the quotient map. Then $U := \pi^{-1}(\overline{U})$ is a submodule of V , so U is finitely generated, say

$$U = Rv_1 + \cdots + Rv_n.$$

Applying π , we get

$$\overline{U} = R(v_1 + W) + \cdots + R(v_n + W).$$

Thus every submodule of V/W is finitely generated, and therefore V/W is Noetherian.

Conversely, assume that both W and V/W are Noetherian. Let U be any submodule of V . We show that U is finitely generated.

First,

$$U \cap W$$

is a submodule of W , so it is finitely generated. Choose generators

$$U \cap W = Rw_1 + \cdots + Rw_m.$$

Next, since

$$(U + W)/W$$

is a submodule of V/W , it is finitely generated as well. Choose elements $u_1, \dots, u_n \in U$ such that

$$(U + W)/W = R(u_1 + W) + \cdots + R(u_n + W).$$

Now let $u \in U$. Since $u + W \in (U + W)/W$, there exist $a_1, \dots, a_n \in R$ such that

$$u + W = \sum_{i=1}^n a_i(u_i + W).$$

Therefore

$$u - \sum_{i=1}^n a_i u_i \in W.$$

Because also $u, u_i \in U$, we have

$$u - \sum_{i=1}^n a_i u_i \in U \cap W.$$

So there exist $b_1, \dots, b_m \in R$ such that

$$u - \sum_{i=1}^n a_i u_i = \sum_{j=1}^m b_j w_j.$$

Hence

$$u = \sum_{i=1}^n a_i u_i + \sum_{j=1}^m b_j w_j.$$

This shows that U is generated by

$$u_1, \dots, u_n, w_1, \dots, w_m.$$

Thus every submodule of V is finitely generated, so V is Noetherian. $\square$

Exercise 5.4.5. Let V be a Noetherian R -module and let $f: V \rightarrow V$ be an epimorphism. Show that f is an R -isomorphism.

Proof. Since f is an epimorphism, it is surjective. It therefore suffices to prove that f is injective.

For each $n \geq 1$, set

$$K_n = \ker(f^n).$$

Then

$$K_1 \subseteq K_2 \subseteq K_3 \subseteq \dots,$$

because if $f^n(x) = 0$, then

$$f^{n+1}(x) = f(f^n(x)) = 0.$$

Since V is Noetherian, this ascending chain stabilizes. Hence there exists $m \geq 1$ such that

$$K_m = K_{m+1}.$$

We claim that $K_m = 0$. Let $x \in K_m$. Because f is surjective, f^m is also surjective, there exists $y \in V$ such that

$$f^m(y) = x.$$

Now

$$f^{2m}(y) = f^m(f^m(y)) = f^m(x) = 0,$$

so $y \in K_{2m} = K_m$. Therefore

$$x = f^m(y) = 0.$$

Thus $K_m = 0$.

Since

$$\ker(f) = K_1 \subseteq K_m = 0,$$

we obtain $\ker(f) = 0$. So f is injective. Being both surjective and injective, f is an isomorphism. $\square$